Using NCBI Virus to prepare data for Selection Analyses

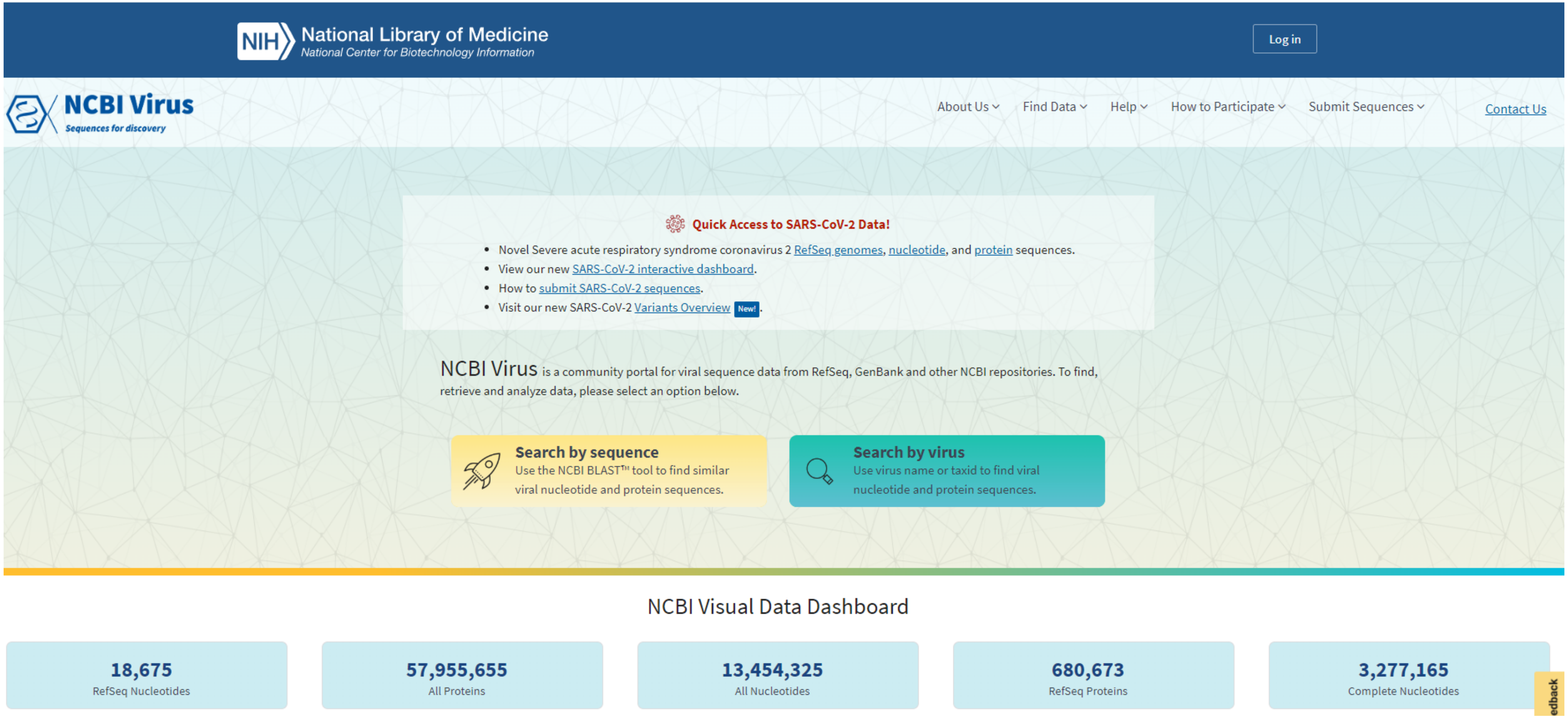
last updated August 5, 2024

Codon alignments and phylogenetic trees

This simple tutorial goes over how to use the browser to access and prepare data for HyPhy-based selection analyses

The database is accessible via: https://www.ncbi.nlm.nih.gov/labs/virus/vssi/#/

Access the NCBI Virus database



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Quick Access to SARS-CoV-2 Data!

- Novel Severe acute respiratory syndrome coronavirus 2 RefSeq genomes, nucleotide, and protein sequences.
- View our new SARS-CoV-2 interactive dashboard.
- How to submit SARS-CoV-2 sequences.
- Visit our new SARS-CoV-2 Variants Overview New!.

NCBI Virus is a community portal for viral sequence data from RefSeq, GenBank and other NCBI repositories. To find, retrieve and analyze data, please select an option below.

Search by sequence
Use the NCBI BLAST™ tool to find similar viral nucleotide and protein sequences.

Search by virus
Use virus name or taxid to find viral nucleotide and protein sequences.

NCBI Visual Data Dashboard

18,675	57,955,655	13,454,325	680,673	3,277,165
RefSeq Nucleotides	All Proteins	All Nucleotides	RefSeq Proteins	Complete Nucleotides

edback

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Search by sequence
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Use virus name or taxid to find viral nucleotide and protein sequences.

Search by virus name or taxonomy

Begin typing a virus name, viral taxonomy group, or taxid to select from a list of suggestions.

virus name or taxid

Popular searches

All viruses | Human viruses | Bacteriophages | New sequences (past one month) | **Up-to-date SARS-CoV-2**

NCBI Visual Data Dashboard

18,675	57,955,655	13,454,325	680,673	3,277,165
RefSeq Nucleotides	All Proteins	All Nucleotides	RefSeq Proteins	Complete Nucleotides

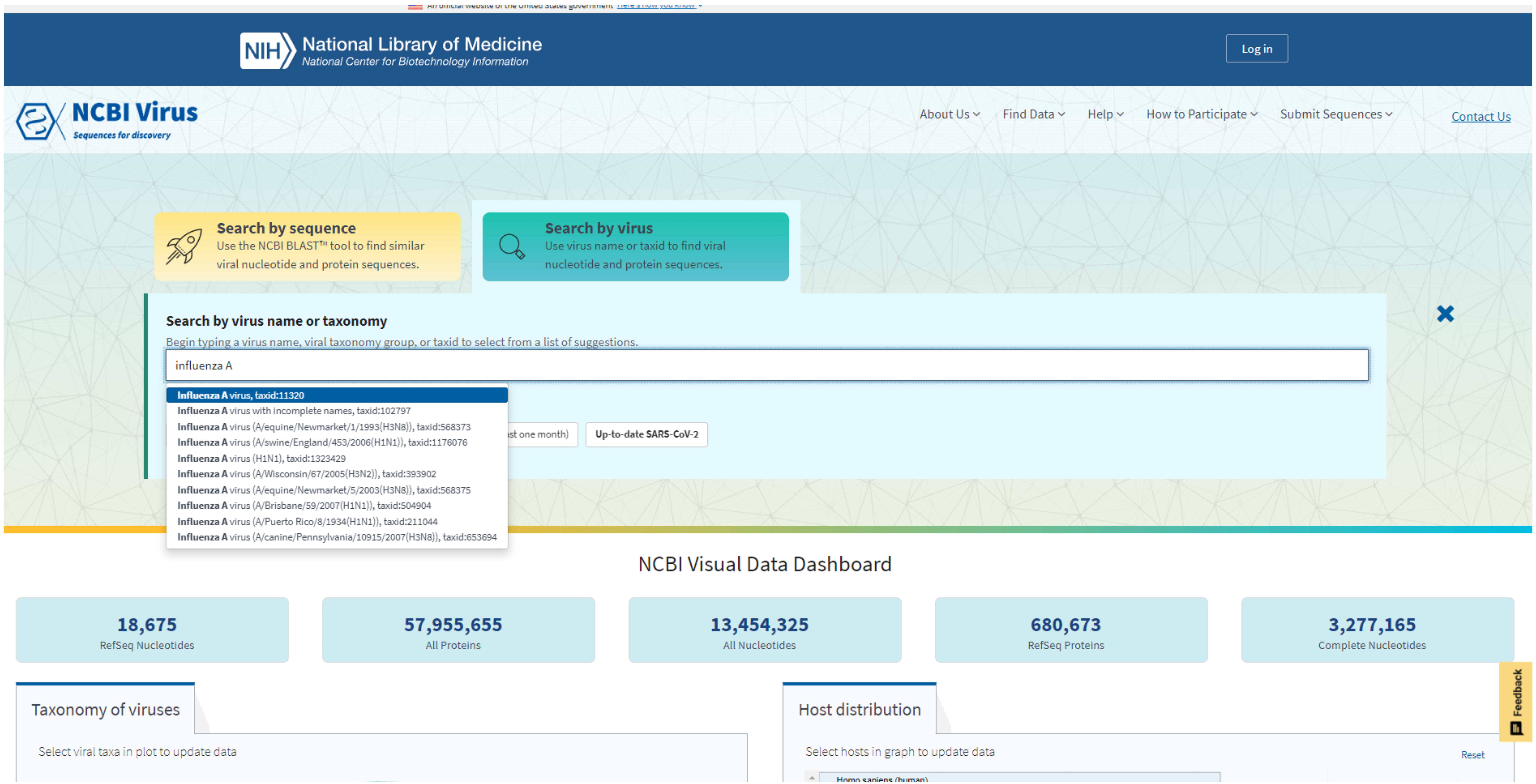
Taxonomy of viruses

Select viral taxa in plot to update data

Host distribution

Select hosts in graph to update data

An Example using Influenza data



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Explore Virus Data

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Popular Searches: **Influenza virus**, Rotavirus, Dengue virus, West Nile virus, Zika virus, MERS coronavirus, Ebolavirus, **SARS-CoV-2 coronavirus**

Advanced Filters for GenBank Sequences | **Visual Filters for GenBank Sequences**

Selected Results: 0 | Align | Build Phylogenetic Tree

New! Randomized subsets in Downloads

You now have the option of downloading a smaller, randomized subset of the data shown in the Results table. Begin by using filters to refine your dataset, select the Nucleotide, Protein, or RefSeq Genome tab above the table for the datatype you would like to download, then follow the prompts in the Download menu. Our Help documentation has more information.

Refine Results — Reset

- Virus: Influenza A virus, taxid:11320
- Accession
- Sequence Length
- Ambiguous Characters
- Sequence Type
- RefSeq Genome Completeness
- Nucleotide Completeness
- Isolate

Applied Filters: Virus (1)

Nucleotide (1,078,897) | **Protein (1,451,218)** | **RefSeq Genome (7)**

Select Columns

Expand Table

Accession	Organism Name	Submitters	Organization	Release Date	Isolate	Species	Length	Nuc Completeness	Geo Location
NC_026431 RefSeq	Influenza A virus (A/Califo…	Garten,R.J., et al.	National Center for Biotec…	2015-02-23		Alphainfluenzavirus influ…	982	complete	USA: California state
NC_026432 RefSeq	Influenza A virus (A/Califo…	Garten,R.J., et al.	National Center for Biotec…	2015-02-23		Alphainfluenzavirus influ…	863	complete	USA: California state
NC_026433 RefSeq	Influenza A virus (A/Califo…	Garten,R.J., et al.	National Center for Biotec…	2015-02-23		Alphainfluenzavirus influ…	1701	complete	USA: California state
NC_026434 RefSeq	Influenza A virus (A/Califo…	Garten,R.J., et al.	National Center for Biotec…	2015-02-23		Alphainfluenzavirus influ…	1410	complete	USA: California state
NC_026435 RefSeq	Influenza A virus (A/Califo…	Shu,B., et al.	National Center for Biotec…	2015-02-23		Alphainfluenzavirus influ…	2274	complete	USA: California sta
NC_026436 RefSeq	Influenza A virus (A/Califo…	Garten,R.J., et al.	National Center for Biotec…	2015-02-23		Alphainfluenzavirus influ…	1497	complete	USA: California sta
NC_026437 RefSeq	Influenza A virus (A/Califo…	Garten,R.J., et al.	National Center for Biotec…	2015-02-23		Alphainfluenzavirus influ…	2151	complete	USA: California state

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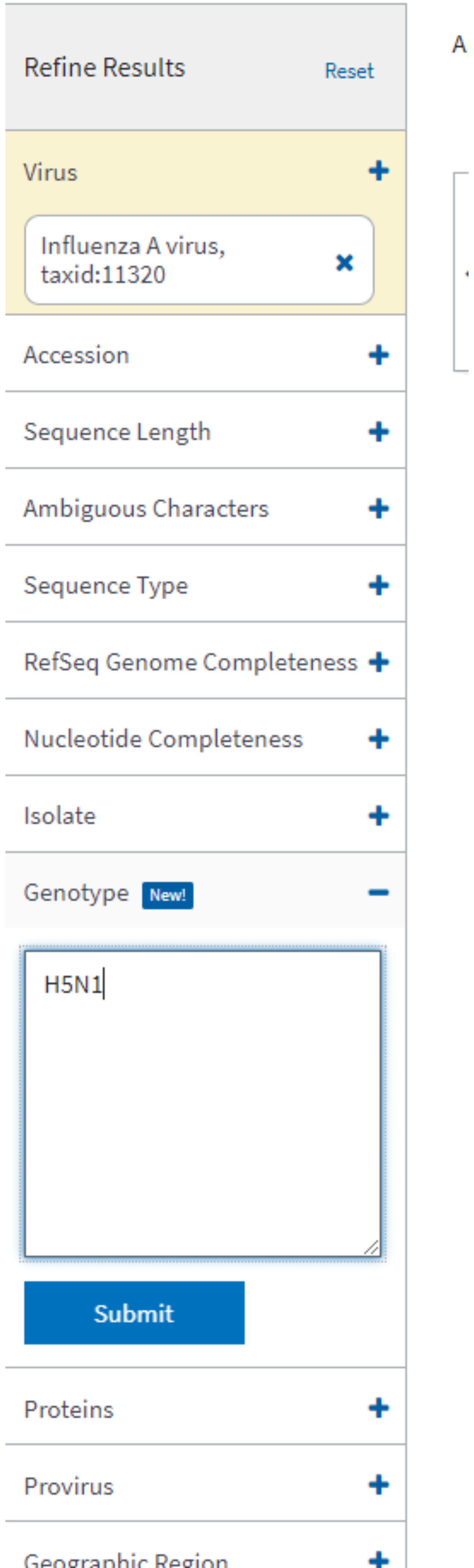
Feedback

Filter search results by Genotype



Refine Results Reset

- Virus +
  - Influenza A virus, taxid:11320 ✖
- Accession +
- Sequence Length +
- Ambiguous Characters +
- Sequence Type +
- RefSeq Genome Completeness +
- Nucleotide Completeness +
- Isolate +
- Genotype New! +
  - H5N1 ✖
- Proteins +
- Provirus +
- Geographic Region +
- Host +
- Submitters +
- Isolation Source +
- Collection Date +

Applied Filters: Virus (1) ✖ Genotype (1) ✖

Nucleotide (54,274) | Protein (74,916) | RefSeq Genome (0)

Select Columns

Expand Table

Accession	Organism Name	Submitters	Organization	Release Date	Isolate	Species	Length	Nuc Completeness	Geo Location
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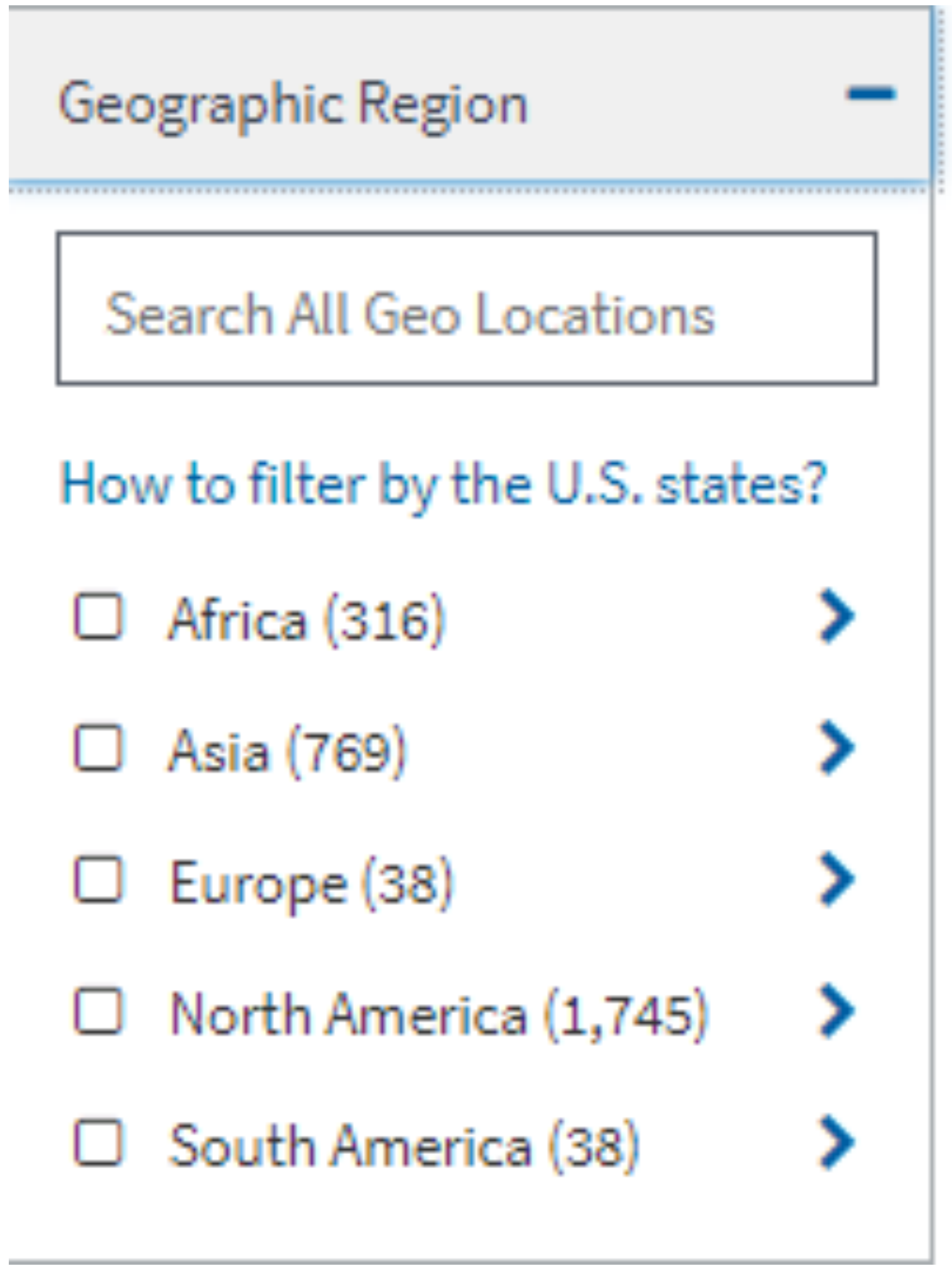
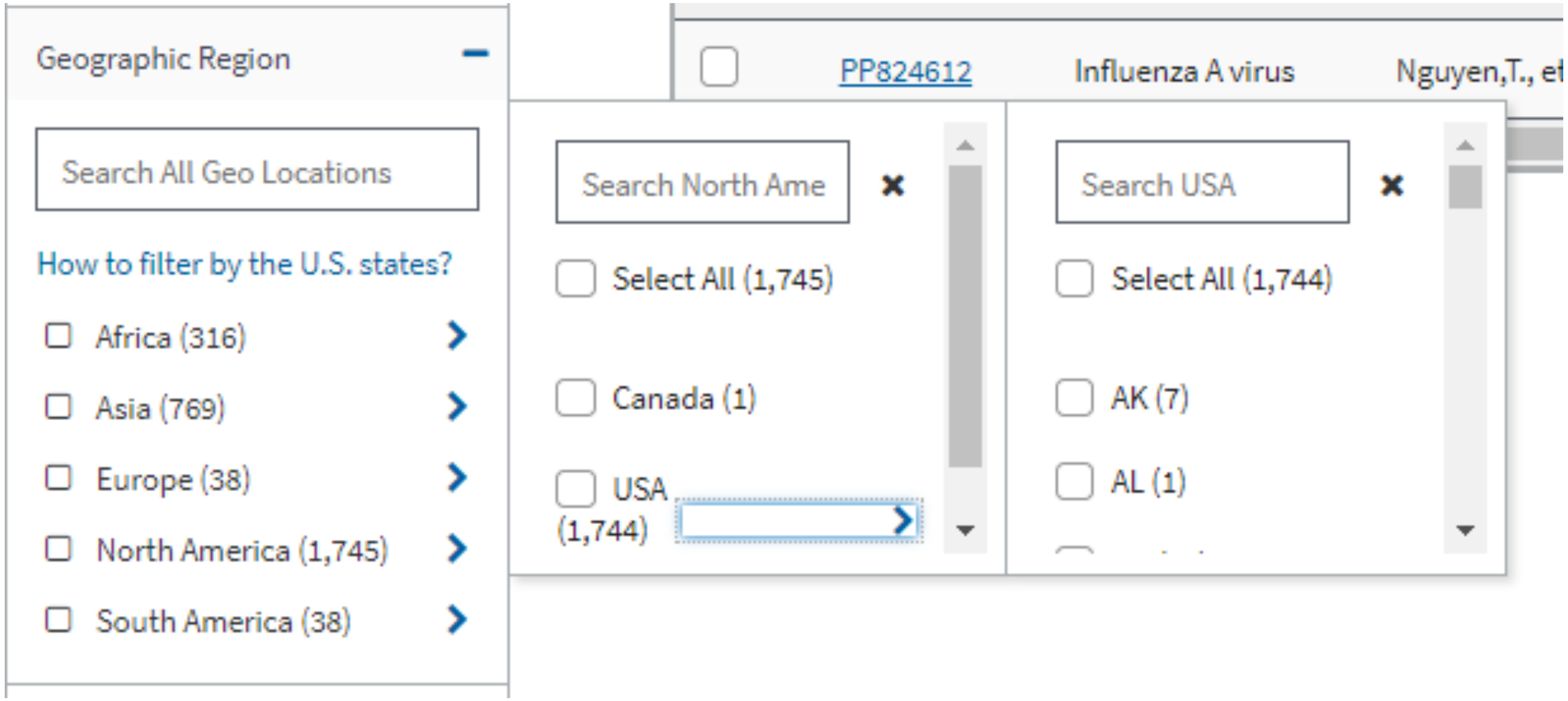
NC_007362 RefSeq	Influenza A virus (A/goose…	Xu,X., et al.	National Center for Biotec…	2005-08-26		Alphainfluenzavirus influe…	1760	complete	
NC_007363 RefSeq	Influenza A virus (A/goose…	Xu,X., et al.	National Center for Biotec…	2005-08-26		Alphainfluenzavirus influe…	1027	complete	
NC_007364 RefSeq	Influenza A virus (A/goose…	Xu,X., et al.	National Center for Biotec…	2005-08-26		Alphainfluenzavirus influe…	865	complete	
PP786397	Influenza A virus	Reischak,D., et…	Ministerio da Agricultura …	2024-06-25	2271-N	Alphainfluenzavirus influe…	2305	partial	Brazil: UbatubaBI
PP786398	Influenza A virus	Reischak,D., et…	Ministerio da Agricultura …	2024-06-25	2271-N	Alphainfluenzavirus influe…	2341	partial	Brazil: UbatubaBI
PP786399	Influenza A virus	Reischak,D., et…	Ministerio da Agricultura …	2024-06-25	2271-N	Alphainfluenzavirus influe…	2233	partial	Brazil: UbatubaBI
PP786400	Influenza A virus	Reischak,D., et…	Ministerio da Agricultura …	2024-06-25	2271-N	Alphainfluenzavirus influe…	1776	partial	Brazil: UbatubaBI
PP786401	Influenza A virus	Reischak,D., et…	Ministerio da Agricultura …	2024-06-25	2271-N	Alphainfluenzavirus influe…	1549	partial	Brazil: UbatubaBI
PP786402	Influenza A virus	Reischak,D., et…	Ministerio da Agricultura …	2024-06-25	2271-N	Alphainfluenzavirus influe…	1447	partial	Brazil: UbatubaBI
PP786403	Influenza A virus	Reischak,D., et…	Ministerio da Agricultura …	2024-06-25	2271-N	Alphainfluenzavirus influe…	1027	partial	Brazil: UbatubaBI
PP786404	Influenza A virus	Reischak,D., et…	Ministerio da Agricultura …	2024-06-25	2271-N	Alphainfluenzavirus influe…	890	partial	Brazil: UbatubaBI
OR839997	Influenza A virus	Reischak,D., et…	MInisterio da Agricultura …	2024-06-24	2108-SN52	Alphainfluenzavirus influe…	1764	partial	Brazil: Bonito
OR839998	Influenza A virus	Reischak,D., et…	MInisterio da Agricultura …	2024-06-24	2108-SN52	Alphainfluenzavirus influe…	1027	partial	Brazil: Bonito

Feedback


Geographic Region
Search All Geo Locations
How to filter by the U.S. states?
Africa (316)
Asia (769)
Europe (38)
North America (1,745)
South America (38)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How to filter by the U.S. states?
Africa (316)
Asia (769)
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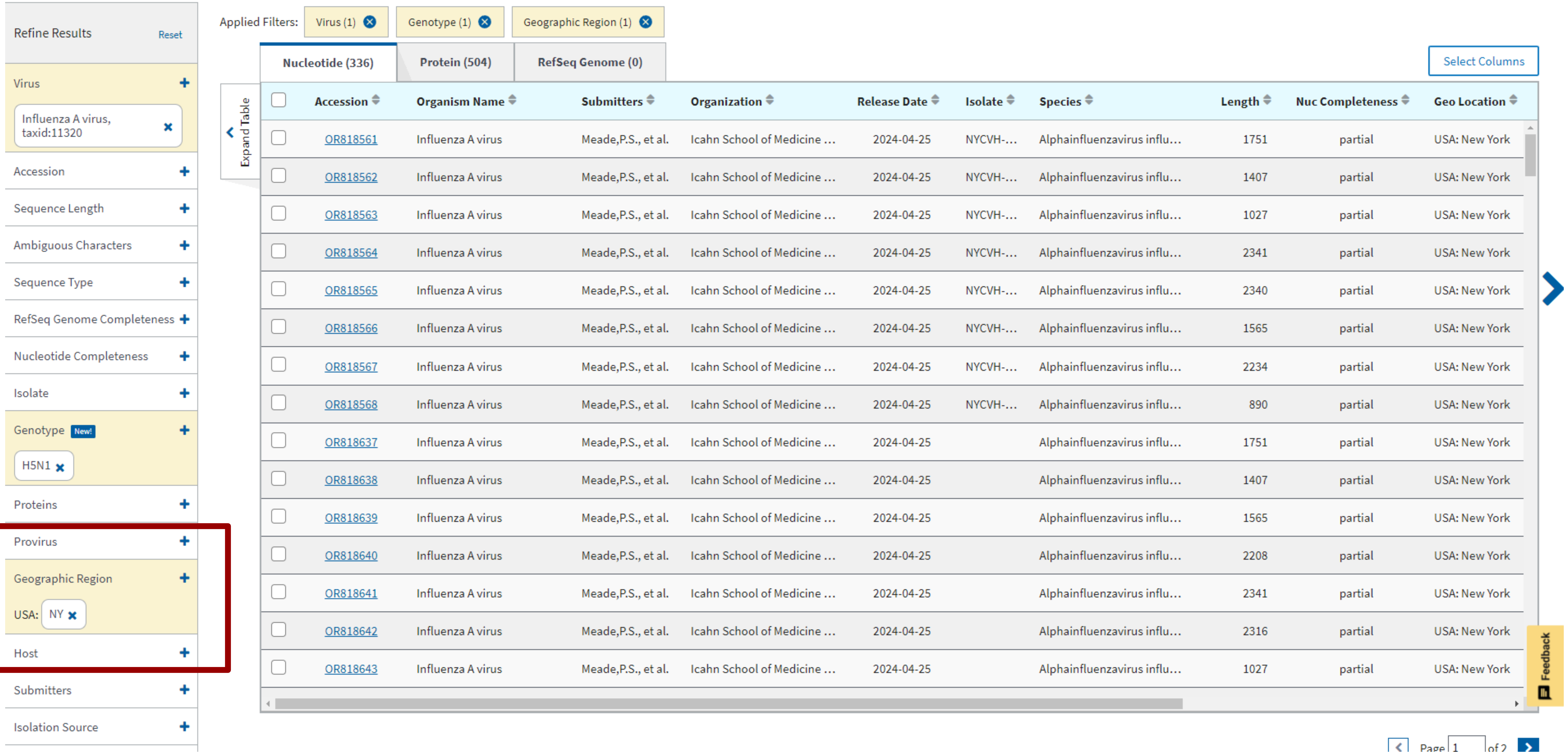
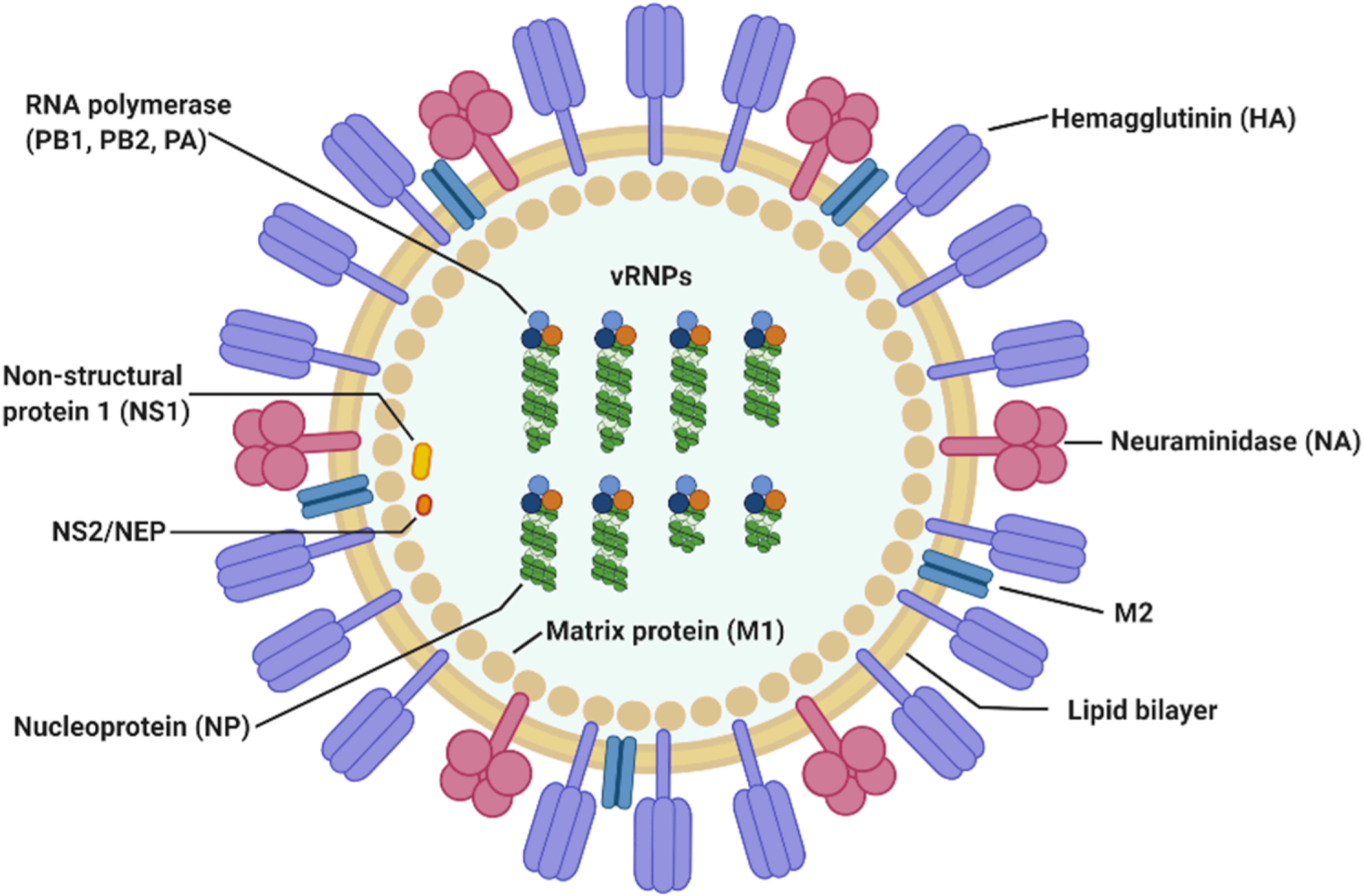
South America (38)
PP824612
Influenza A virus
Search North Ame
Select All (1,745)
Canada (1)
USA (1,744)
Search USA
Select All (1,744)
AK (7)
AL (1)

Filter by Geographic region



	Accession	Organism Name	Submitters	Organization	Release Date	Isolate	Species	Length	Nuc Completeness	Geo Location
	OR818561	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-...	Alphainfluenzavirus influ...	1751	partial	USA: New York
	OR818562	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-...	Alphainfluenzavirus influ...	1407	partial	USA: New York
	OR818563	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-...	Alphainfluenzavirus influ...	1027	partial	USA: New York
	OR818564	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-...	Alphainfluenzavirus influ...	2341	partial	USA: New York
	OR818565	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-...	Alphainfluenzavirus influ...	2340	partial	USA: New York
	OR818566	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-...	Alphainfluenzavirus influ...	1565	partial	USA: New York
	OR818567	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-...	Alphainfluenzavirus influ...	2234	partial	USA: New York
	OR818568	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-...	Alphainfluenzavirus influ...	890	partial	USA: New York
	OR818637	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	1751	partial	USA: New York
	OR818638	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	1407	partial	USA: New York
	OR818639	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	1565	partial	USA: New York
	OR818640	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	2208	partial	USA: New York
	OR818641	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	2341	partial	USA: New York
	OR818642	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	2316	partial	USA: New York
	OR818643	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	1027	partial	USA: New York


RNA polymerase
(PB1, PB2, PA)
Hemagglutinin (HA)
vRNPs
Non-structural
protein 1 (NS1)
Neuraminidase (NA)
NS2/NEP
M2
Matrix protein (M1)
Lipid bilayer
Nucleoprotein (NP)

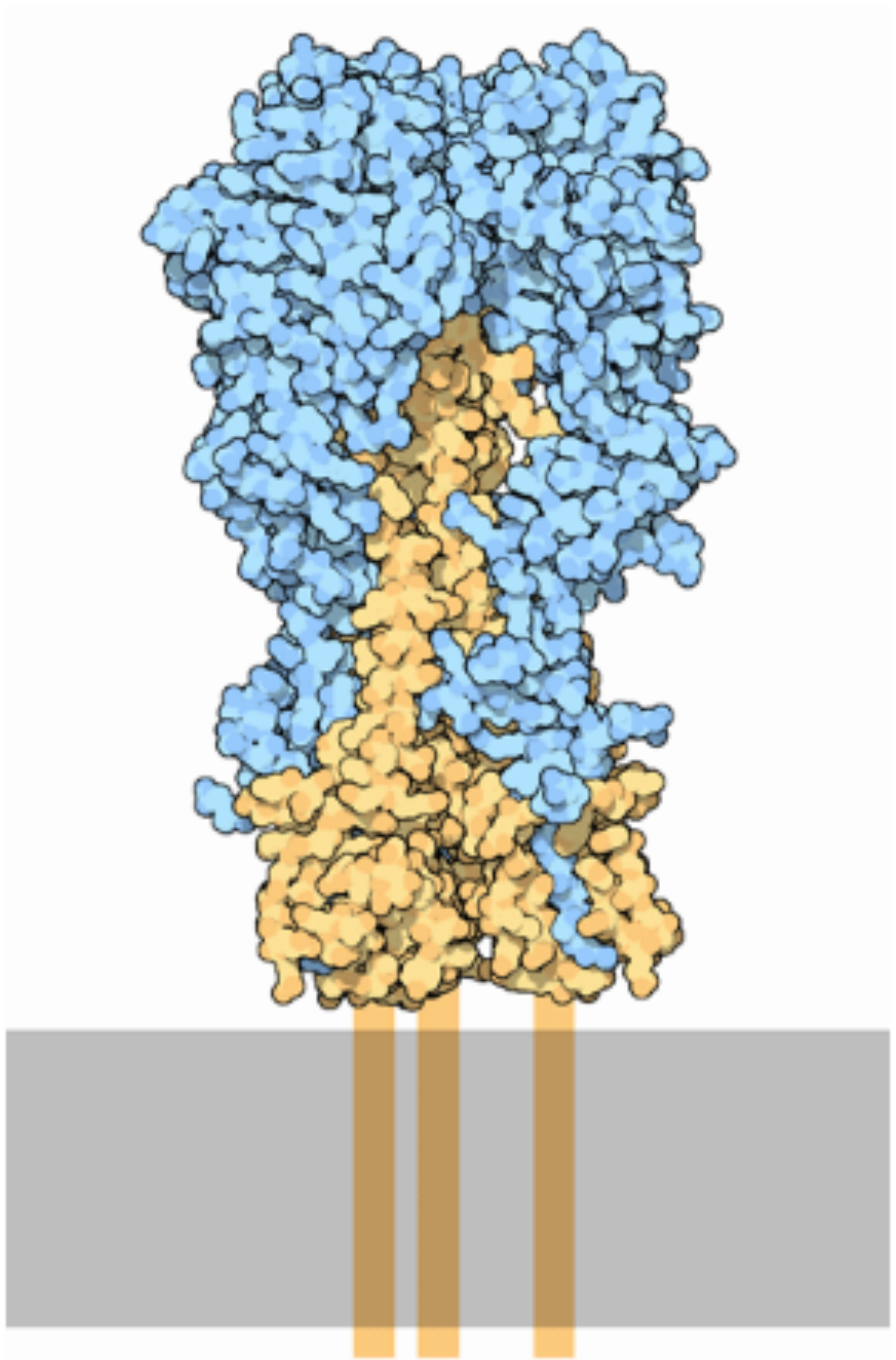
Molecule of the Month: Hemagglutinin

Influenza virus binds to cells and infects them using hemagglutinin

Influenza virus is a dangerous enemy. Normally, the immune system fights off infections, eradicating the viruses and causing a few days of miserable flu symptoms. Yearly flu vaccines prime our immune system, making it ready to fight the most common strains of influenza virus. But once every couple of decades, a new strain of influenza appears that is far more pathogenic, allowing it to spread rapidly. This happened at the end of World War I, and the resultant pandemic killed over 20 million people, more than twice the number of people that were killed in the war.

Target and Attack

Hemagglutinin is one of the reasons that influenza virus is so effective. It is a spike-shaped protein that extends from the surface of the virus. In the active form, shown here from PDB entry 1ruz, hemagglutinin is composed of two different types of chains, shown in blue and orange. The blue chains are the targeting mechanism: they search for specific sugar chains on our cellular proteins. When they find the proper one, hemagglutinin binds to the cell and the orange chains initiate the attack, as shown on the next page. The name hemagglutinin refers to the ability of influenza to agglutinate red blood cells: the virus is covered with many hemagglutinin molecules, which together can glue many red blood cells together into a visible clump.



Influenza hemagglutinin. The portion spanning the membrane is not included in the structure and is shown schematically.

Download high quality TIFF image

Stealthy Subtypes

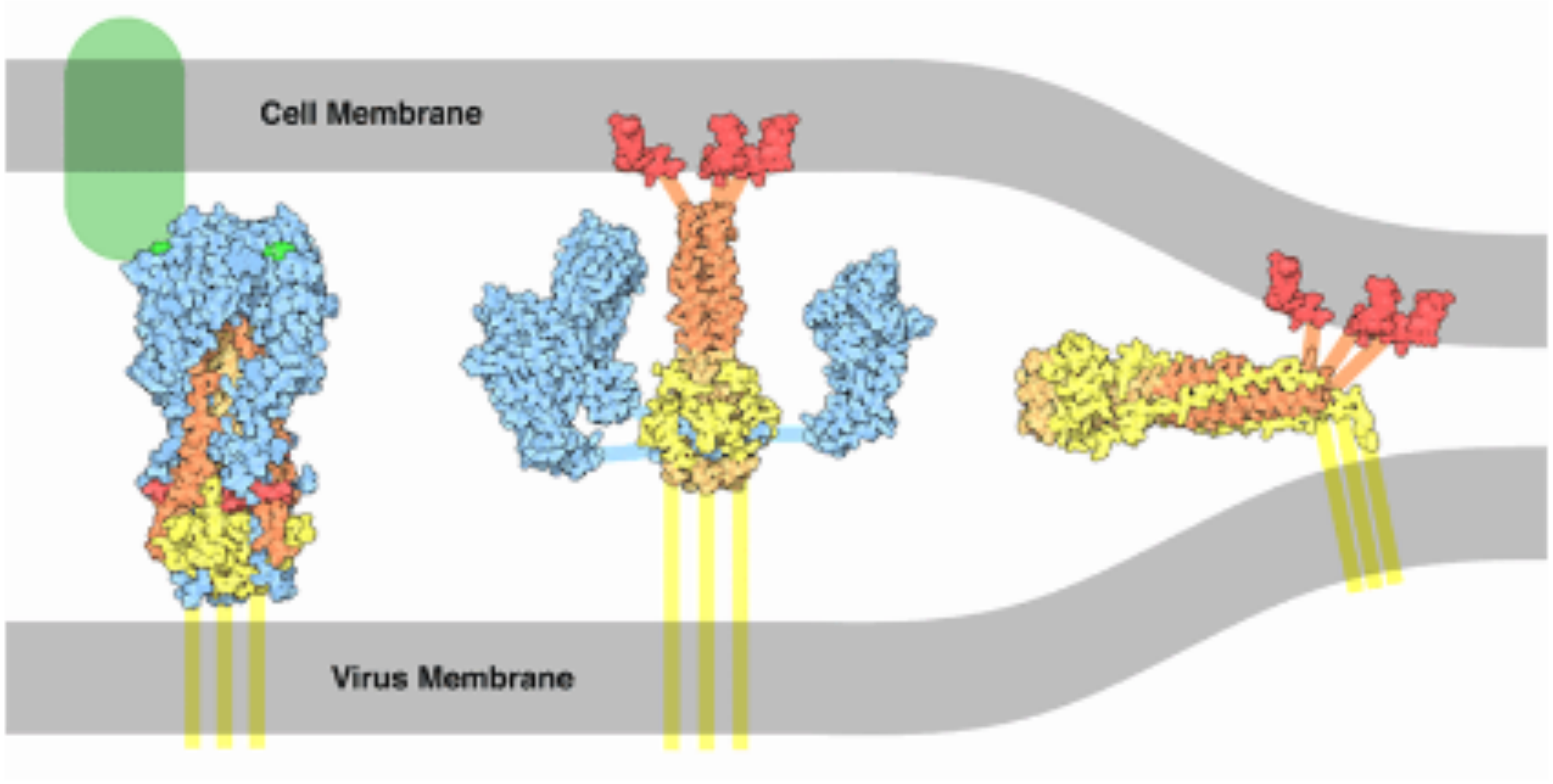
The specificity, and thus the danger, of each strain of influenza virus depends on the particular type hemagglutinin that it carries. Over a dozen subtypes of hemagglutinin are known. Three of these (termed H1, H2, and H3) attack humans--they specialize in finding the particular sugars in our respiratory tract, so the infection occurs there when we get the flu. Other subtypes, such as H5, attack proteins in the digestive system of birds. Most of these are not dangerous to us, and do not typically threaten the lives of the birds, so they exist as an invisible reservoir of virus. A potential danger occurs when these different strains start trading genes.

The H5N1 avian influenza virus that is showing up in the news is decimating bird populations, but it is not currently a danger to us since it doesn't have the right hemagglutinin to attack human cells (the N1 designation refers to the subtype of a second virus protein: neuraminidase). However, there is the possibility that it could acquire a human-specific hemagglutinin, and then it could cause real problems. One way for this to happen involves pigs. Pigs are susceptible to viruses of several subtypes, both the types that attack birds and the types that attack us. If a single pig gets infected with two different viruses at the same time, the viruses can shuffle and trade genes during the infection. This could potentially be a way to construct a virus with the virulence of the bird virus, combined with the ability to attack human cells.

Deadly Agent

The hemagglutinin shown here is taken from an actual virus of the pandemic that killed so many people in 1918. The DNA encoding this hemagglutinin was isolated from preserved samples, and the hemagglutinin was made in the laboratory according to this genetic information. Two crystal structures were obtained, the active form shown here (PDB entry 1ruz), and a precursor form in PDB entry 1rd8. The protein is tethered to the virus membrane by a short segment of protein that is not seen in the crystal structure and is shown schematically here at the bottom.

https://pdb101.rcsb.org/motm/76



Conformational changes in hemagglutinin that lead to membrane fusion.

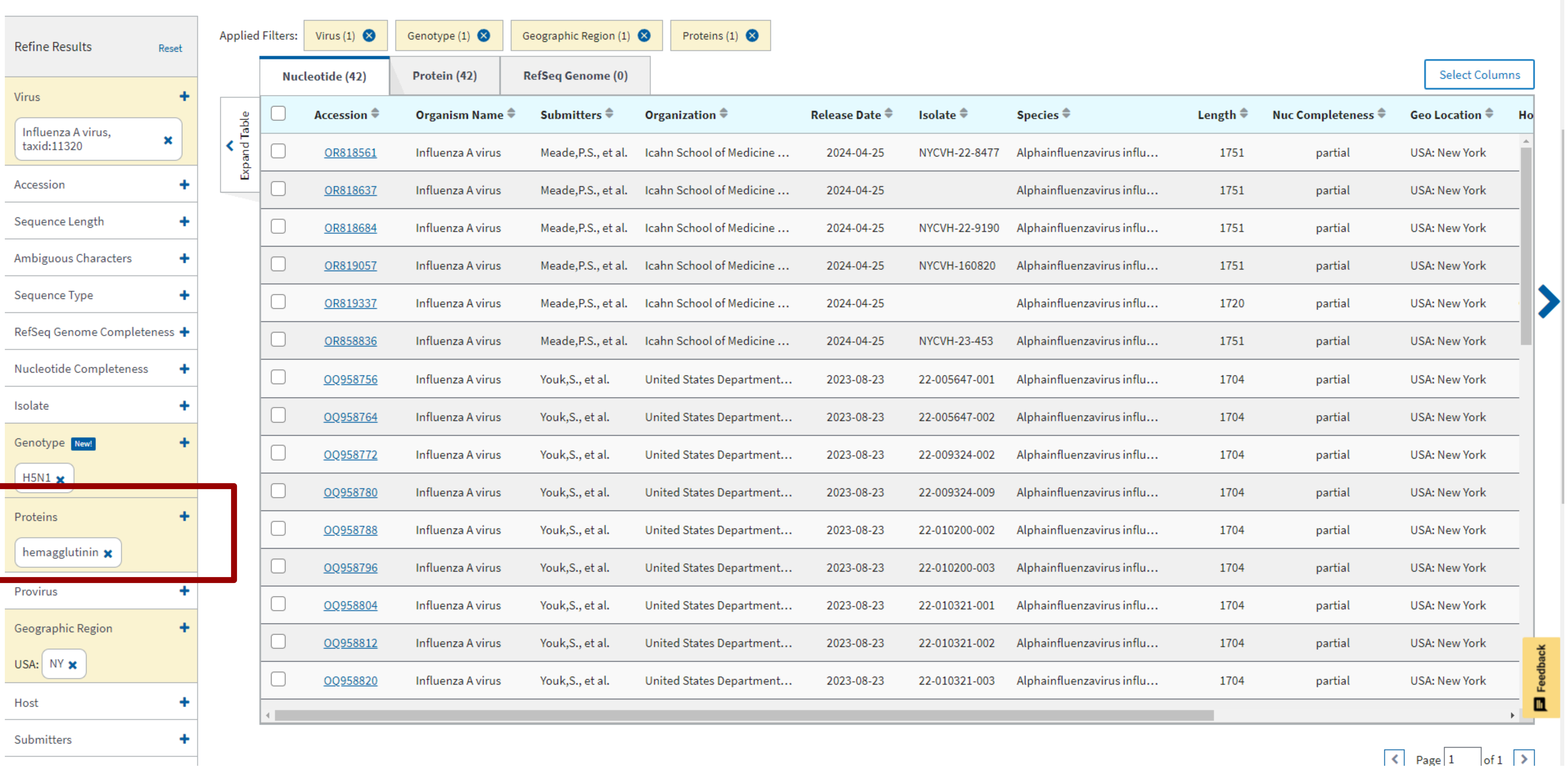
Download high quality TIFF image

Hemagglutinin in Action

Hemagglutinin is a deadly molecular machine that targets and attacks cells. This occurs in several steps. First, the three binding sites at the top of the spike bind to sugars on cellular proteins, shown in green at the top left (PDB entry 1hge). Then, the whole virus is carried inside the cell into the endosome and the cell adds acid, which normally digests the stuff inside the endosome. But in the case of the virus, the acidic environment serves to arm the attack mechanism. In acid, hemagglutinin unfolds and then refolds into an entirely different shape. The portions shown in orange and red are normally folded against the protein, but in acid, they pop out and point upward, as shown in the center illustration (PDB entries 1htm, 1ibn and 2vir). The red portion, termed the fusion peptide, has a strong affinity for membranes, so it inserts into the cell membrane and locks the virus to the cell. Then, as shown on the left (PDB entry 1qu1), the yellow portions zip up the side of the protein, pulling the two membranes close together. Finally, the new conformation of hemagglutinin somehow causes the two membranes to fuse--that part is still not well understood--and the viral RNA flows into the cell, starting the process of infection.

https://pdb101.rcsb.org/motm/76

Filter by viral protein



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OR818637	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	1751	partial	USA: New York	
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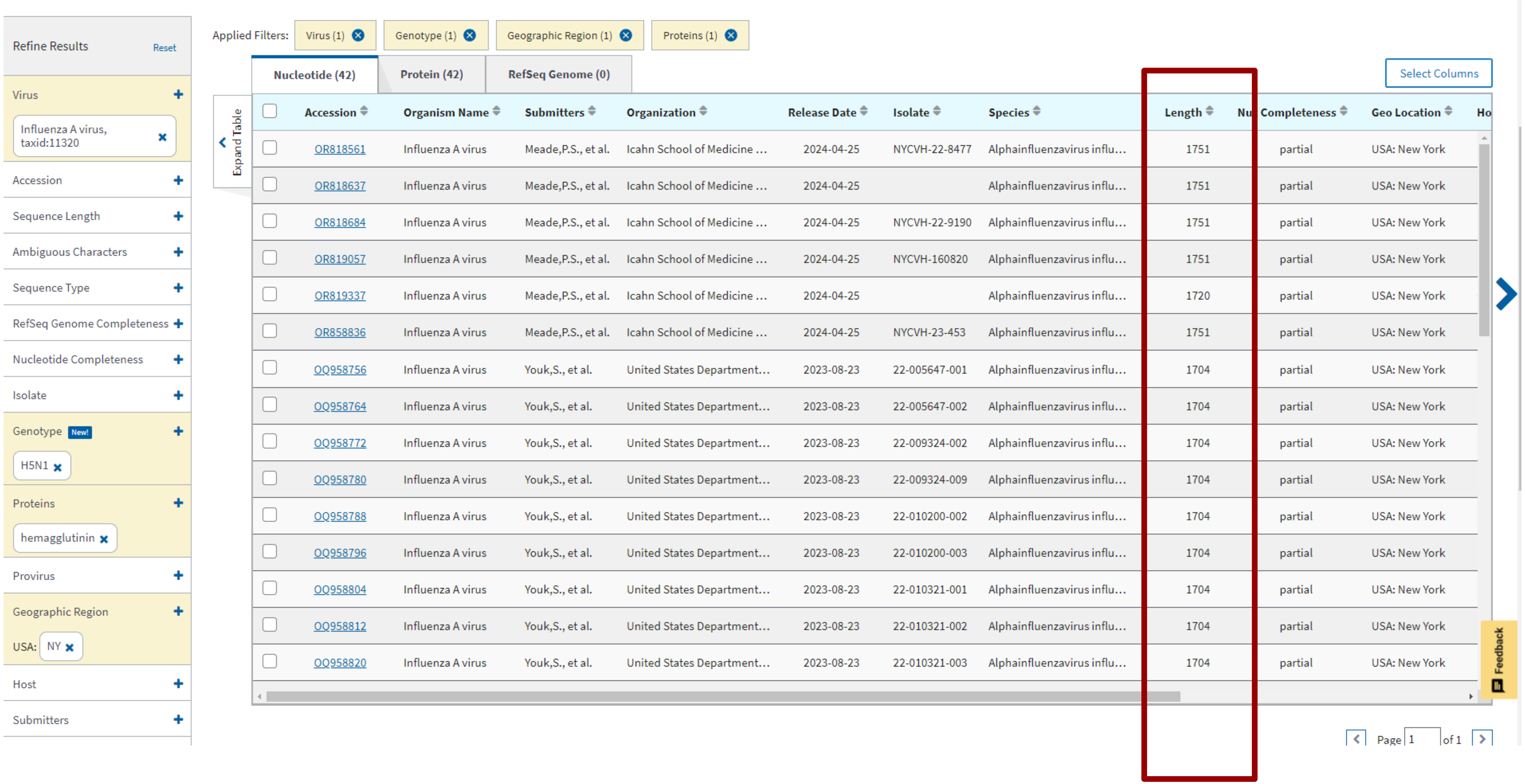
OR819057	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-160820	Alphainfluenzavirus influ...	1751	partial	USA: New York	
OR819337	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	1720	partial	USA: New York	
OR858836	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-23-453	Alphainfluenzavirus influ...	1751	partial	USA: New York	
OQ958756	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-005647-001	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958764	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-005647-002	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958772	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-009324-002	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958780	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-009324-009	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958788	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010200-002	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958796	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010200-003	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958804	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010321-001	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958812	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010321-002	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958820	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010321-003	Alphainfluenzavirus influ...	1704	partial	USA: New York	

Filter by protein length



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OR818561	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-22-8477	Alphainfluenzavirus influ...	1751		partial	USA: New York	
OR818637	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	1751		partial	USA: New York	
OR818684	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-22-9190	Alphainfluenzavirus influ...	1751		partial	USA: New York	
OR819057	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-160820	Alphainfluenzavirus influ...	1751		partial	USA: New York	
OR819337	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25		Alphainfluenzavirus influ...	1720		partial	USA: New York	
OR858836	Influenza A virus	Meade,P.S., et al.	Icahn School of Medicine ...	2024-04-25	NYCVH-23-453	Alphainfluenzavirus influ...	1751		partial	USA: New York	
OQ958756	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-005647-001	Alphainfluenzavirus influ...	1704		partial	USA: New York	
OQ958764	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-005647-002	Alphainfluenzavirus influ...	1704		partial	USA: New York	
OQ958772	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-009324-002	Alphainfluenzavirus influ...	1704		partial	USA: New York	
OQ958780	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-009324-009	Alphainfluenzavirus influ...	1704		partial	USA: New York	
OQ958788	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010200-002	Alphainfluenzavirus influ...	1704		partial	USA: New York	
OQ958796	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010200-003	Alphainfluenzavirus influ...	1704		partial	USA: New York	
OQ958804	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010321-001	Alphainfluenzavirus influ...	1704		partial	USA: New York	
OQ958812	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010321-002	Alphainfluenzavirus influ...	1704		partial	USA: New York	
OQ958820	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010321-003	Alphainfluenzavirus influ...	1704		partial	USA: New York	

Refine Results Reset

Virus +

Influenza A virus, taxid:11320 ✖

Accession +

Sequence Length +

Min: 1704 **Max:** 1704 ✖

Ambiguous Characters +

Sequence Type +

RefSeq Genome Completeness +

Nucleotide Completeness +

Isolate +

Genotype New! +

H5N1 ✖

Proteins +

hemagglutinin ✖

Provirus +

Geographic Region +

Applied Filters: Virus (1) ⊗ Genotype (1) ⊗ Geographic Region (1) ⊗ Proteins (1) ⊗ Sequence Length (1) ⊗

Nucleotide (36) | **Protein (0)** | **RefSeq Genome (0)**

Select Columns

Expand Table

Organism Name	Submitters	Organization	Release Date	Isolate	Species	Length	Nuc Completeness	Geo Location	Host
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-005647-001	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-005647-002	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-009324-002	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-009324-009	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-010200-002	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-010200-003	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-010321-001	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-010321-002	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-010321-003	Alphainfluenzavirus influ…	1704	partial	USA: New York	
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-009119-001	Alphainfluenzavirus influ…	1704	partial	USA: New York	Sibirionetta formosa
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-010803-001	Alphainfluenzavirus influ…	1704	partial	USA: New York	Haliaeetus leucocephalus
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-012660-001	Alphainfluenzavirus influ…	1704	partial	USA: New York	Haliaeetus leucocephalus
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-012661-001	Alphainfluenzavirus influ…	1704	partial	USA: New York	Haliaeetus leucocephalus
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-013006-001	Alphainfluenzavirus influ…	1704	partial	USA: New York	Haliaeetus leucocephalus
Influenza A virus	Youk,S., et al.	United States Department…	2023-08-23	22-009396-001	Alphainfluenzavirus influ…	1704	partial	USA: New York	Branta canadensis

Feedback

Now that the sequence selection is done

We can Align and quality-control our sequences

We can Infer a phylogenetic tree


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Zika virus
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Ebolavirus
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Visual Filters for GenBank Sequences
Selected Results: 36
Align
Build Phylogenetic Tree
New! Randomized subsets in Downloads
You now have the option of downloading a smaller, randomized subset of the data shown in the Results table. Begin by using filters to refine your dataset, select the Nucleotide, Protein, or RefSeq Genome tab above the table for the datatype you would like to download, then follow the prompts in the Download menu. Our Help documentation has more information.
Refine Results
Reset
Virus
Influenza A virus, taxid:11320
Accession
Sequence Length
Min: 1704
Max: 1704
Ambiguous Characters
Sequence Type
RefSeq Genome Completeness
Nucleotide Completeness
Applied Filters:
Virus (1)
Genotype (1)
Geographic Region (1)
Proteins (1)
Sequence Length (1)
Nucleotide (36)
Protein (0)
RefSeq Genome (0)
Select Columns
Expand Table
Accession
Organism Name
Submitters
Organization
Release Date
Isolate
Species
Length
Nuc Completeness
Geo Location
OQ958756 Influenza A virus Youk,S., et al. United States Department... 2023-08-23 22-005647-001 Alphainfluenzavirus influ... 1704 partial USA: New York
OQ958764 Influenza A virus Youk,S., et al. United States Department... 2023-08-23 22-005647-002 Alphainfluenzavirus influ... 1704 partial USA: New York
OQ958772 Influenza A virus Youk,S., et al. United States Department... 2023-08-23 22-009324-002 Alphainfluenzavirus influ... 1704 partial USA: New York
OQ958780 Influenza A virus Youk,S., et al. United States Department... 2023-08-23 22-009324-009 Alphainfluenzavirus influ... 1704 partial USA: New York
OQ958788 Influenza A virus Youk,S., et al. United States Department... 2023-08-23 22-010200-002 Alphainfluenzavirus influ... 1704 partial USA: New York
OQ958796 Influenza A virus Youk,S., et al. United States Department... 2023-08-23 22-010200-003 Alphainfluenzavirus influ... 1704 partial USA: New York
OQ958804 Influenza A virus Youk,S., et al. United States Department... 2023-08-23 22-010321-001 Alphainfluenzavirus influ... 1704 partial USA: New York
OQ958812 Influenza A virus Youk,S., et al. United States Department... 2023-08-23 22-010321-002 Alphainfluenzavirus influ... 1704 partial USA: New York
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OQ963804.1
OQ958820.1
OQ958812.1
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OQ959548.1
OQ962524.1
OQ959556.1
OQ963036.1
OQ963044.1
OQ959540.1
Influenza A virus
DNA: 1 - 1,704 (1,704 bases shown) - anchor consensus
Rows shown: 36/
Feedback

Explore your data (Important!)

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OQ963804.1
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OQ962516.1
OQ960876.1
OQ959548.1
OQ962524.1
OQ959556.1
OQ963036.1
OQ963044.1
OQ959540.1
Influenza A virus
DNA: 524 - 736 (213 bases shown) - anchor consensus
Rows shown: 36/3
edback

Where do these sequences come from?

Multiple Alignment

Influenza A virus (A/Wood Duck/New York/22-011188-001/2022(H5N1)) segment 4 hemagglutinin (HA) gene, complete cds

GenBank: OQ967076.1

FASTA Graphics

Go to:

```
LOCUS       OQ967076                1704 bp    cRNA    linear   VRL 23-AUG-2023
DEFINITION  Influenza A virus (A/Wood Duck/New York/22-011188-001/2022(H5N1))
            segment 4 hemagglutinin (HA) gene, complete cds.
ACCESSION   OQ967076
VERSION     OQ967076.1
KEYWORDS    .
SOURCE      Influenza A virus
  ORGANISM  Influenza A virus
            Viruses; Riboviria; Orthornavirae; Negarnaviricota;
            Polyploviricotina; Insthoviricetes; Articulavirales;
            Orthomyxoviridae; Alphainfluenzavirus; Alphainfluenzavirus
            influenzae.
REFERENCE   1  (bases 1 to 1704)
  AUTHORS   Youk,S., Torchetti,M.K., Lantz,K., Lenoch,J.B., Killian,M.L.,
            Leyson,C., Bevins,S.N., Dilione,K., Ip,H.S., Stallknecht,D.E.,
            Poulson,R.L., Suarez,D.L., Swayne,D.E. and Pantin-Jackwood,M.J.
  TITLE     H5N1 highly pathogenic avian influenza clade 2.3.4.4b in wild and
            domestic birds: Introductions into the United States and
            reassortments, December 2021-April 2022
  JOURNAL   Virology 587, 109860 (2023)
```

Influenza A virus (A/Snow Goose/New York/22-008835-002/2022(H5N1)) segment 4 hemagglutinin (HA) gene, complete cds

GenBank: OQ964204.1

FASTA Graphics

Go to:

```
LOCUS       OQ964204                1704 bp    cRNA    linear   VRL 23-AUG-2023
DEFINITION  Influenza A virus (A/Snow Goose/New York/22-008835-002/2022(H5N1))
            segment 4 hemagglutinin (HA) gene, complete cds.
ACCESSION   OQ964204
VERSION     OQ964204.1
KEYWORDS    .
SOURCE      Influenza A virus
  ORGANISM  Influenza A virus
            Viruses; Riboviria; Orthornavirae; Negarnaviricota;
            Polyploviricotina; Insthoviricetes; Articulavirales;
            Orthomyxoviridae; Alphainfluenzavirus; Alphainfluenzavirus
            influenzae.
REFERENCE   1  (bases 1 to 1704)
  AUTHORS   Youk,S., Torchetti,M.K., Lantz,K., Lenoch,J.B., Killian,M.L.,
            Leyson,C., Bevins,S.N., Dilione,K., Ip,H.S., Stallknecht,D.E.,
            Poulson,R.L., Suarez,D.L., Swayne,D.E. and Pantin-Jackwood,M.J.
  TITLE     H5N1 highly pathogenic avian influenza clade 2.3.4.4b in wild and
            domestic birds: Introductions into the United States and
            reassortments, December 2021-April 2022
  JOURNAL   Virology 587, 109860 (2023)
```

Influenza A virus (A/Backyard bird/New York/22-010200-002/2022(H5N1)) segment 4 hemagglutinin (HA) gene, complete cds

GenBank: OQ958788.1

FASTA Graphics

Go to:

```
LOCUS       OQ958788                1704 bp    cRNA    linear   VRL 23-AUG-2023
DEFINITION  Influenza A virus (A/Backyard bird/New
            York/22-010200-002/2022(H5N1)) segment 4 hemagglutinin (HA) gene,
            complete cds.
ACCESSION   OQ958788
VERSION     OQ958788.1
KEYWORDS    .
SOURCE      Influenza A virus
  ORGANISM  Influenza A virus
            Viruses; Riboviria; Orthornavirae; Negarnaviricota;
            Polyploviricotina; Insthoviricetes; Articulavirales;
            Orthomyxoviridae; Alphainfluenzavirus; Alphainfluenzavirus
            influenzae.
REFERENCE   1  (bases 1 to 1704)
  AUTHORS   Youk,S., Torchetti,M.K., Lantz,K., Lenoch,J.B., Killian,M.L.,
            Leyson,C., Bevins,S.N., Dilione,K., Ip,H.S., Stallknecht,D.E.,
            Poulson,R.L., Suarez,D.L., Swayne,D.E. and Pantin-Jackwood,M.J.
  TITLE     H5N1 highly pathogenic avian influenza clade 2.3.4.4b in wild and
            domestic birds: Introductions into the United States and
            reassortments, December 2021-April 2022
  JOURNAL   Virology 587, 109860 (2023)
   PUBMED   37572517
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Influenza A virus (A/Cooper's Hawk/New York/22-010033-001/2022(H5N1)) segment 4 hemagglutinin (HA) gene, complete cds

GenBank: OQ960876.1

FASTA Graphics

Go to:

```
LOCUS       OQ960876                1704 bp    cRNA    linear   VRL 23-AUG-2023
DEFINITION  Influenza A virus (A/Cooper's Hawk/New
            York/22-010033-001/2022(H5N1)) segment 4 hemagglutinin (HA) gene,
            complete cds.
ACCESSION   OQ960876
VERSION     OQ960876.1
KEYWORDS    .
SOURCE      Influenza A virus
  ORGANISM  Influenza A virus
            Viruses; Riboviria; Orthornavirae; Negarnaviricota;
            Polyploviricotina; Insthoviricetes; Articulavirales;
            Orthomyxoviridae; Alphainfluenzavirus; Alphainfluenzavirus
            influenzae.
REFERENCE   1  (bases 1 to 1704)
  AUTHORS   Youk,S., Torchetti,M.K., Lantz,K., Lenoch,J.B., Killian,M.L.,
            Leyson,C., Bevins,S.N., Dilione,K., Ip,H.S., Stallknecht,D.E.,
            Poulson,R.L., Suarez,D.L., Swayne,D.E. and Pantin-Jackwood,M.J.
  TITLE     H5N1 highly pathogenic avian influenza clade 2.3.4.4b in wild and
            domestic birds: Introductions into the United States and
            reassortments, December 2021-April 2022
  JOURNAL   Virology 587, 109860 (2023)
```


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OQ959548.1
OQ962524.1
OQ959556.1
OQ963036.1
OQ963044.1
OQ959540.1
Influenza A virus
DNA: 1 - 1,704 (1,704 bases shown) - anchor consensus
Rows shown: 36/
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AliView

About

AliView is yet another alignment viewer and editor, but this is probably one of the fastest and most intuitive to use, not so bloated and hopefully to your liking.

The general idea when designing this program has always been **usability** and **speed**, all new functions are optimized so they do not affect the general performance and capability to work swiftly with large alignments. The speed in rendering even makes it possible to work with large alignments on older hardware.

A need to easily sort, view, remove, edit and merge sequences from large transcriptome datasets initiated the design of the program.

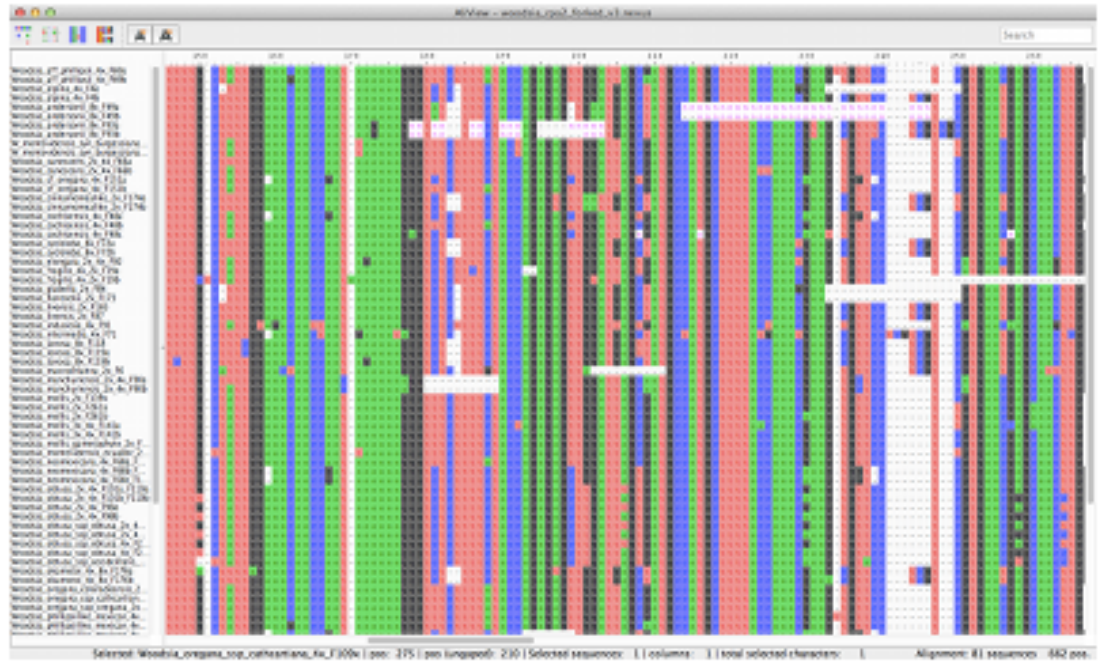
It is of course also working very well with smaller datasets:)

The program is developed at the department of Systematic Biology, Uppsala University, so there is probably a predominance in functionality supporting those working with phylogenies.

Citation: Larsson, A. (2014). AliView: a fast and lightweight alignment viewer and editor for large data sets. *Bioinformatics*30(22): 3276-3278. http://dx.doi.org/10.1093/bioinformatics/btu531

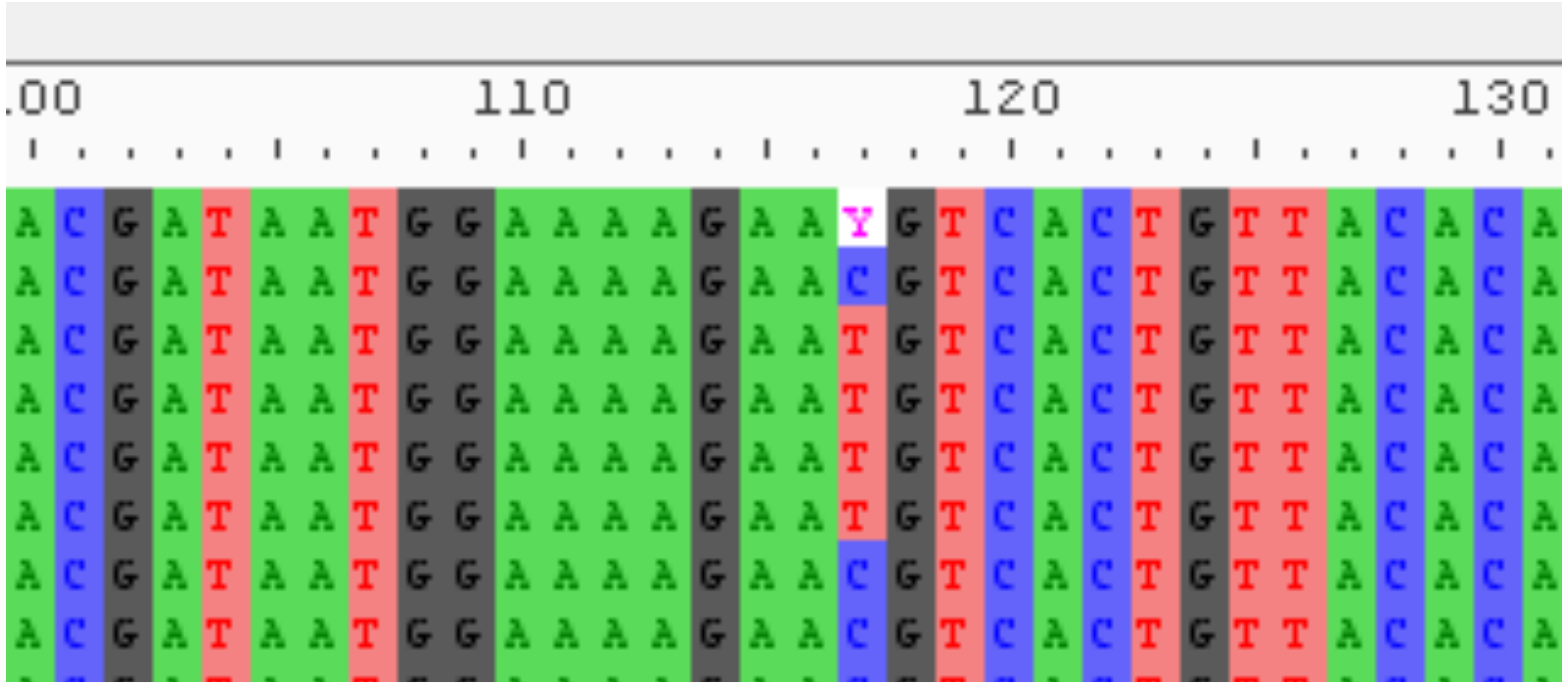
The program is released under the **Open Source** Software License 'GNU General Public License, version 3.0 (GPLv3)'
Source code is available at **GitHub:** https://github.com/AliView/AliView

Download the latest stable version: 1.28 (24/Nov/2021) (Mac OS X, Windows, Linux)



Download at https://ormbunkar.se/aliview/

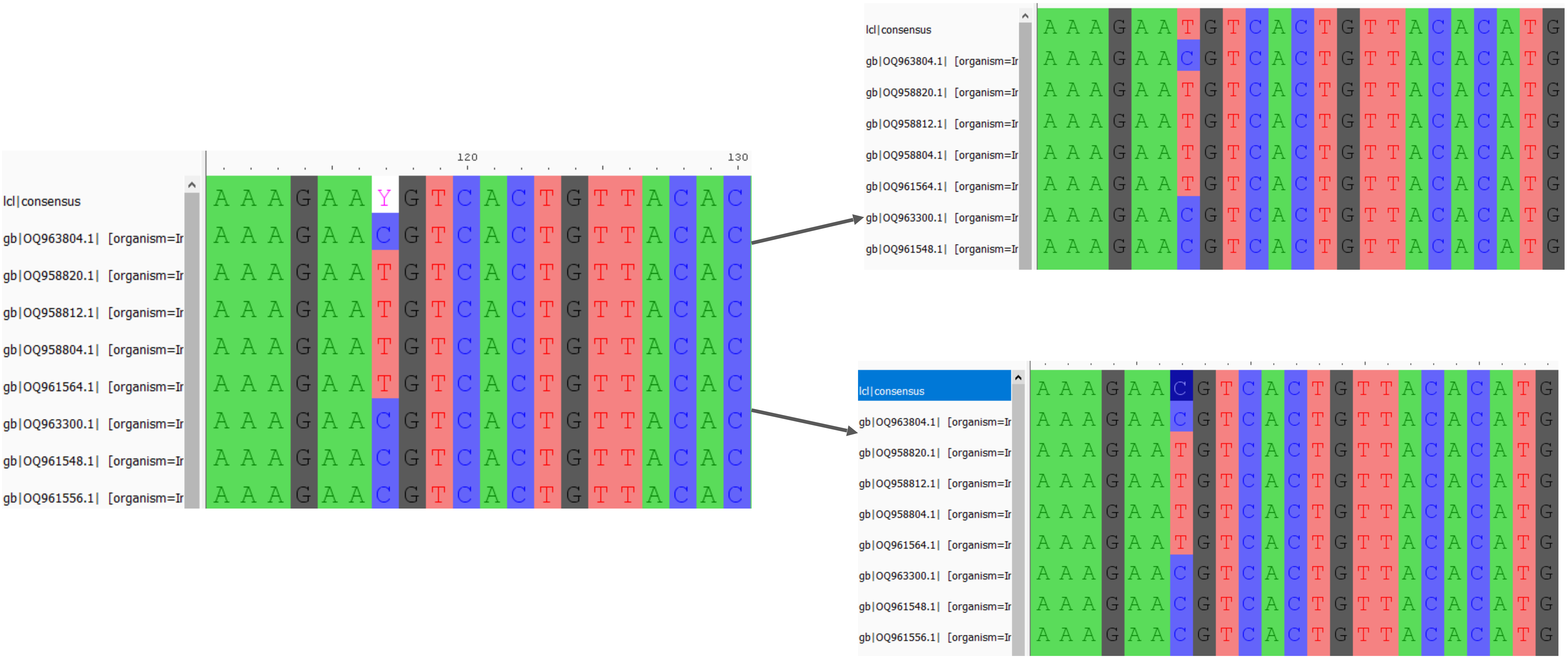
Ambiguous bases can disrupt selection analyses



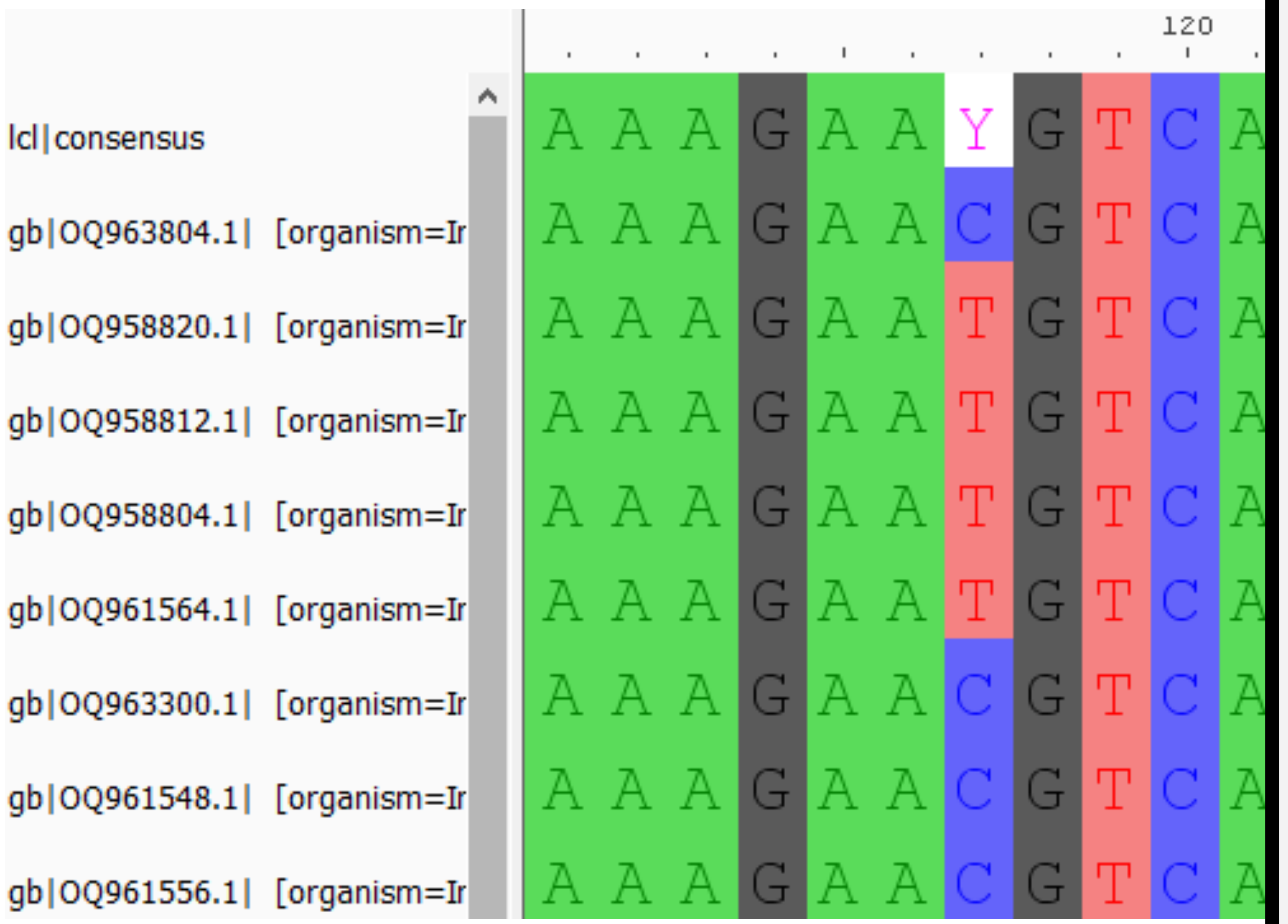
Ambiguous

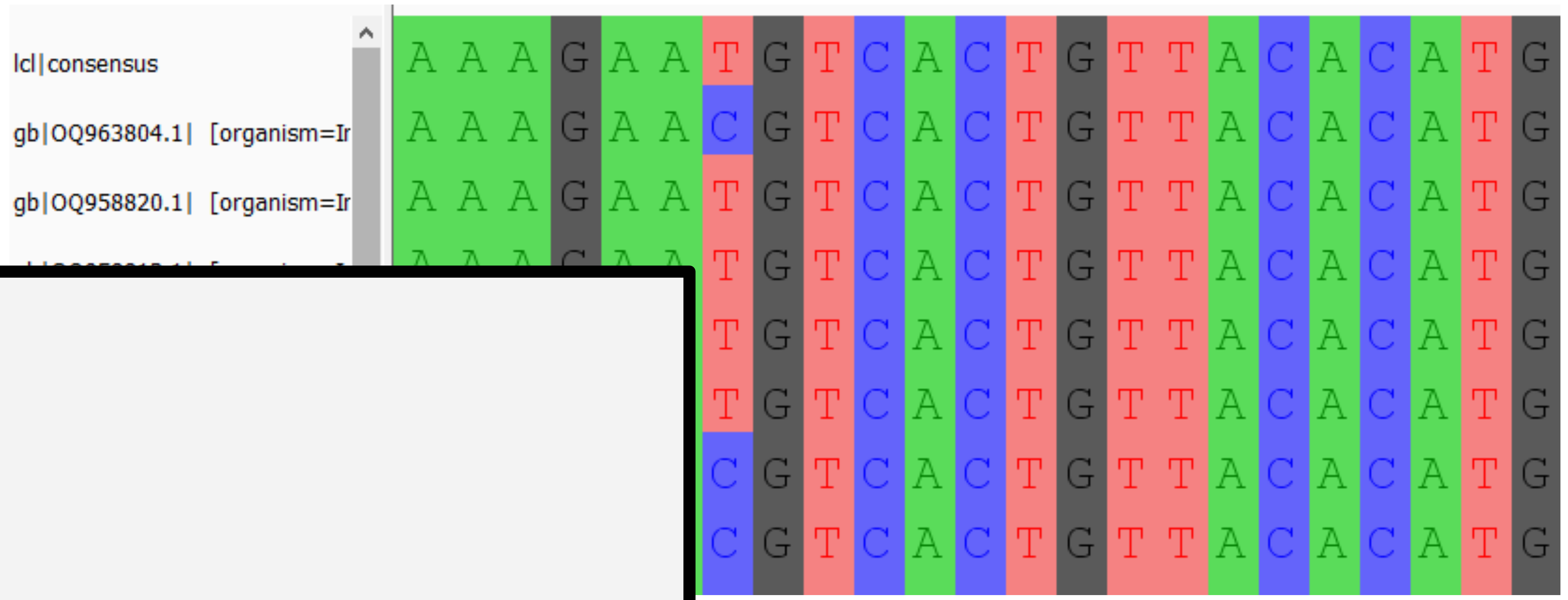
- **N** Any nucleotide (A or C or G or T or U)
- **R** Purine (A or G)
- **Y** Pyrimidine (T or C)
- **K** Keto (G or T)
- **M** Amino (A or C)
- **S** Strong interaction (3 H bonds) (G or C)
- **W** Weak interaction (2 H bonds) (A or T)
- **B** Not A (C or G or T)
- **D** Not C (A or G or T)
- **H** Not G (A or C or T)
- **V** Not T or U (A or C or G)

Methods to resolve this: Convert Y to (C or T)

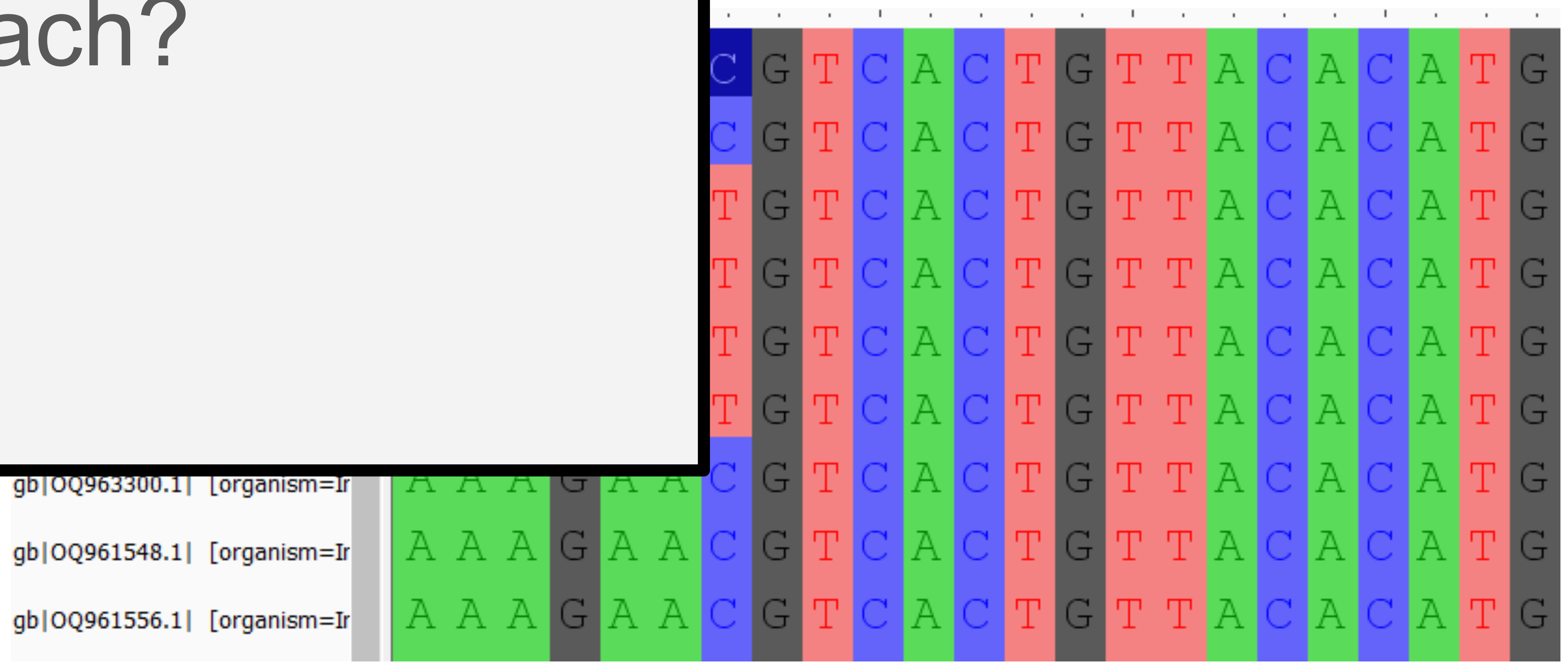


Methods to resolve this: Convert Y to (C or T)

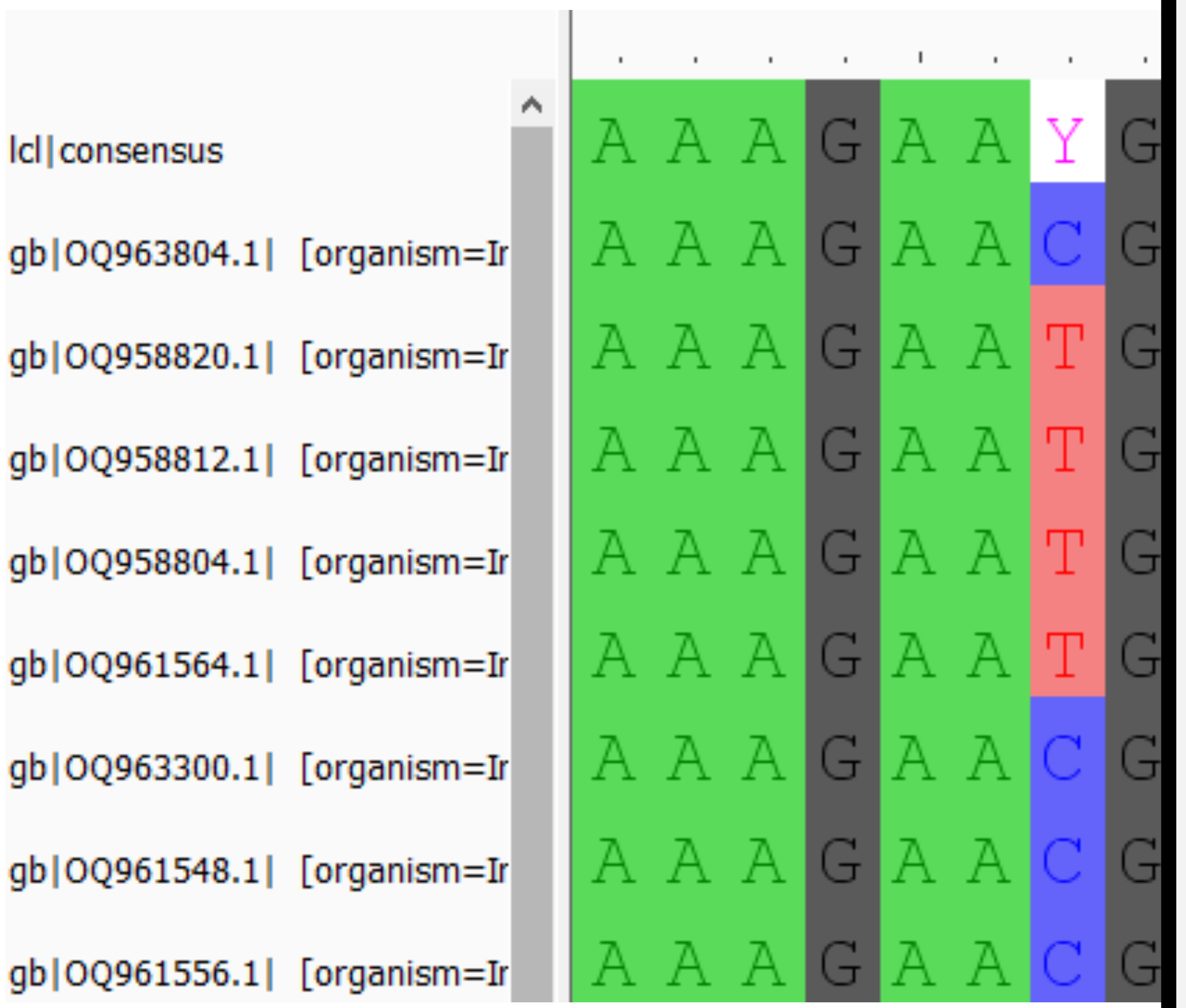




What is the problem with this approach?



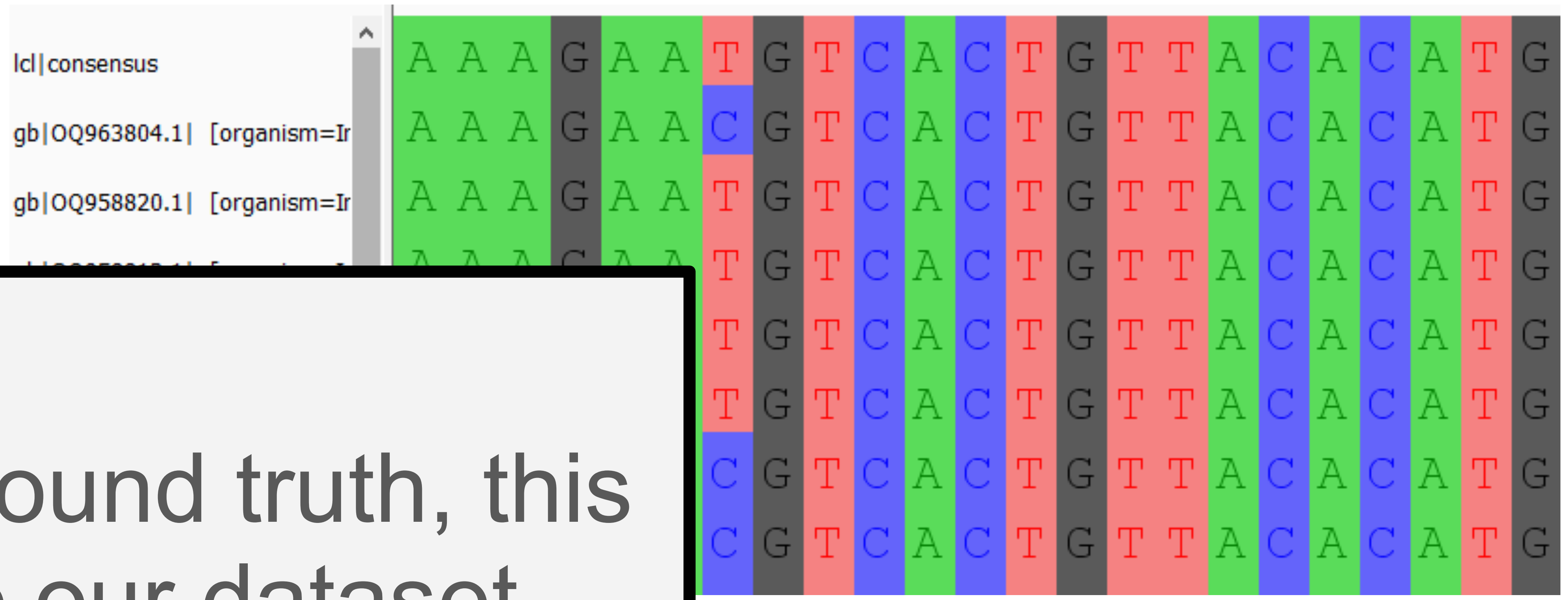
Methods to resolve this: Convert Y to (C or T)

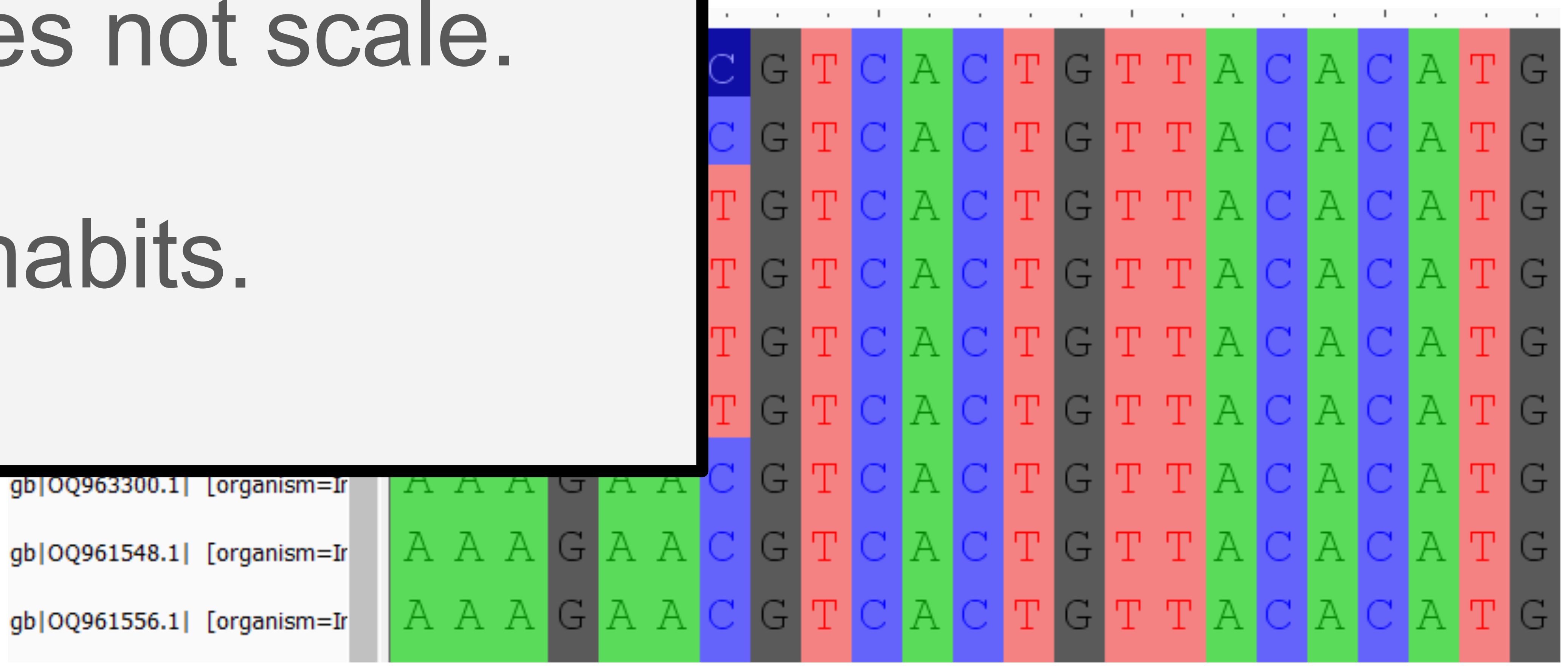


We don't know the ground truth, this introduces bias into our dataset.

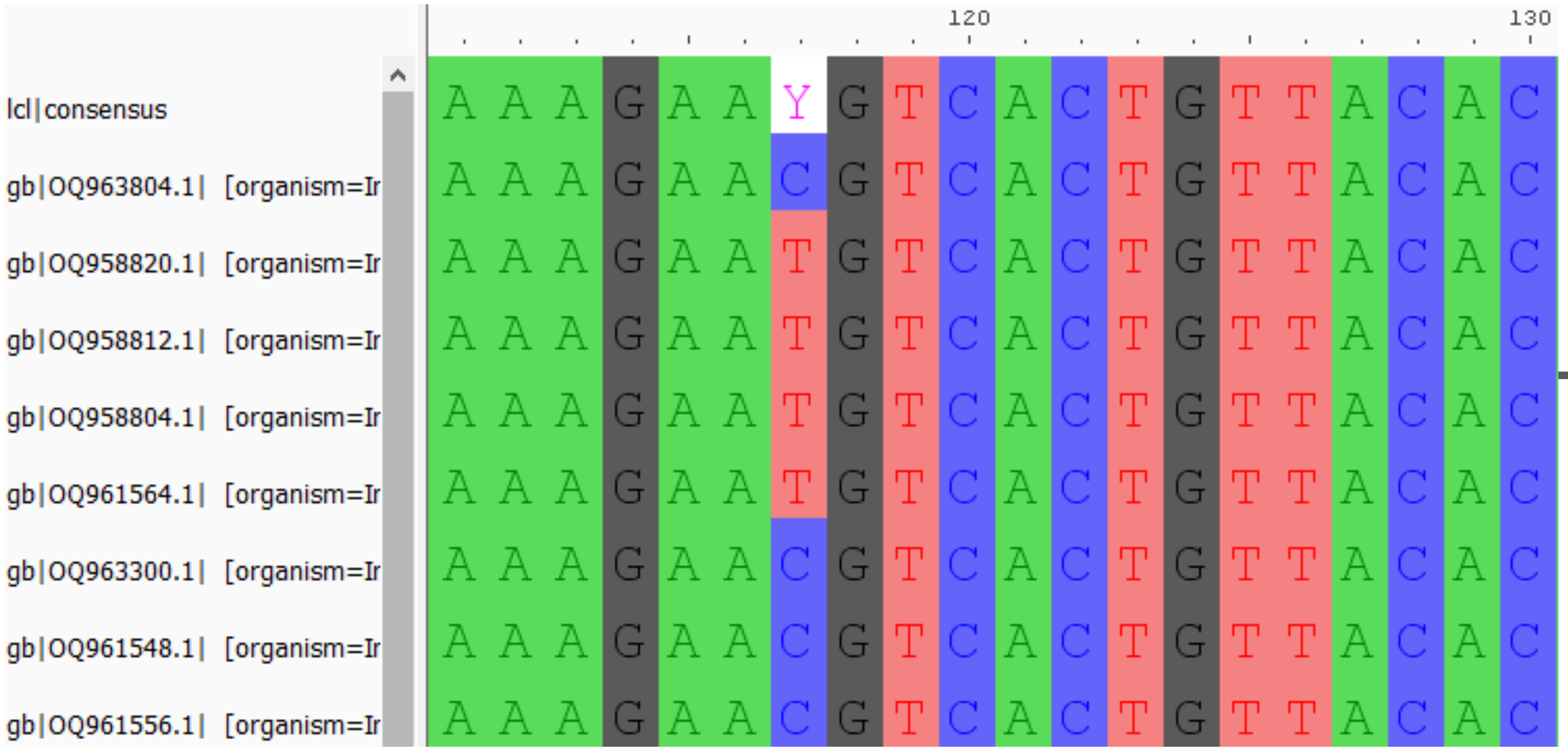
This approach does not scale.

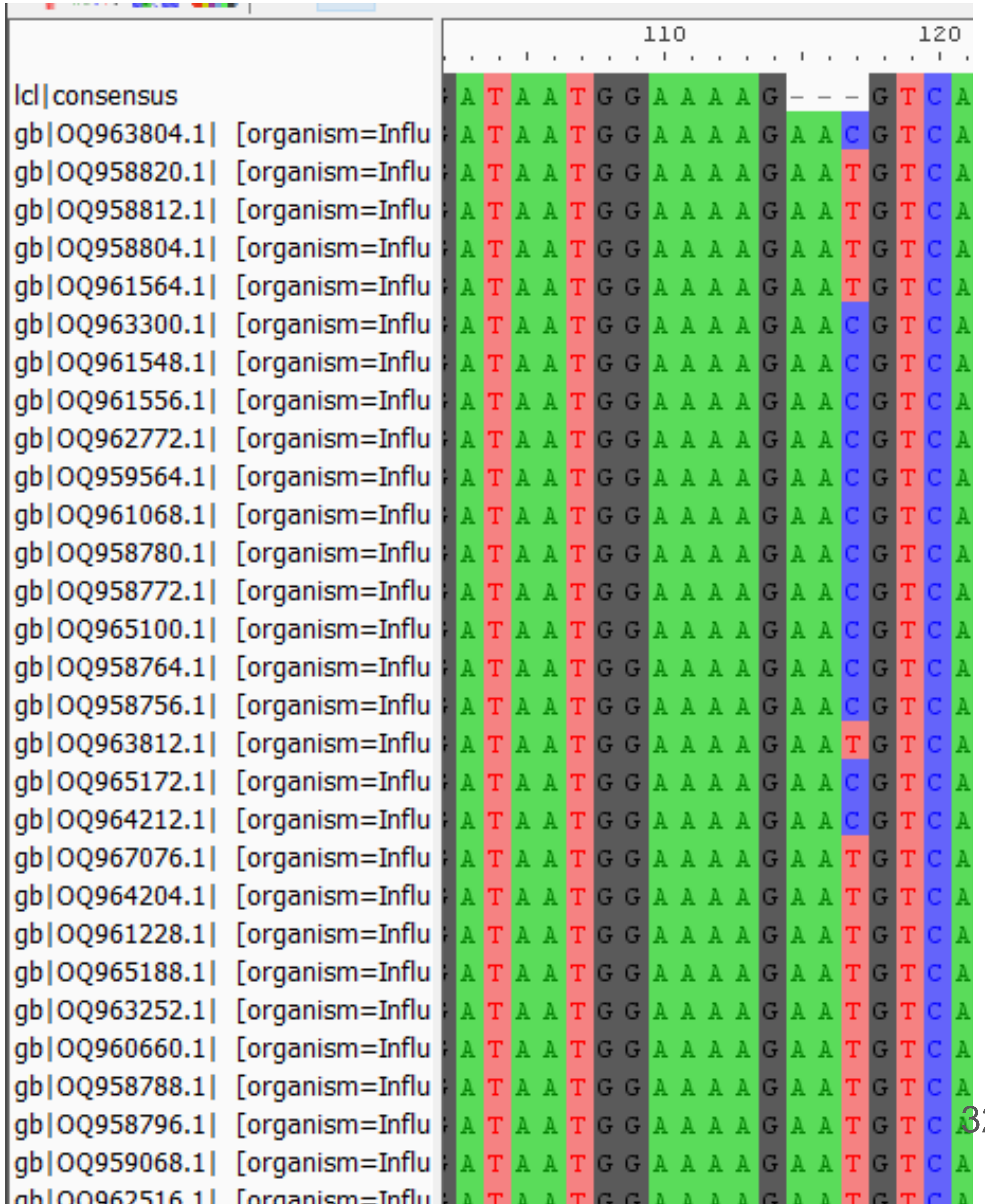
Builds bad habits.





Methods to resolve this: Mask the codon

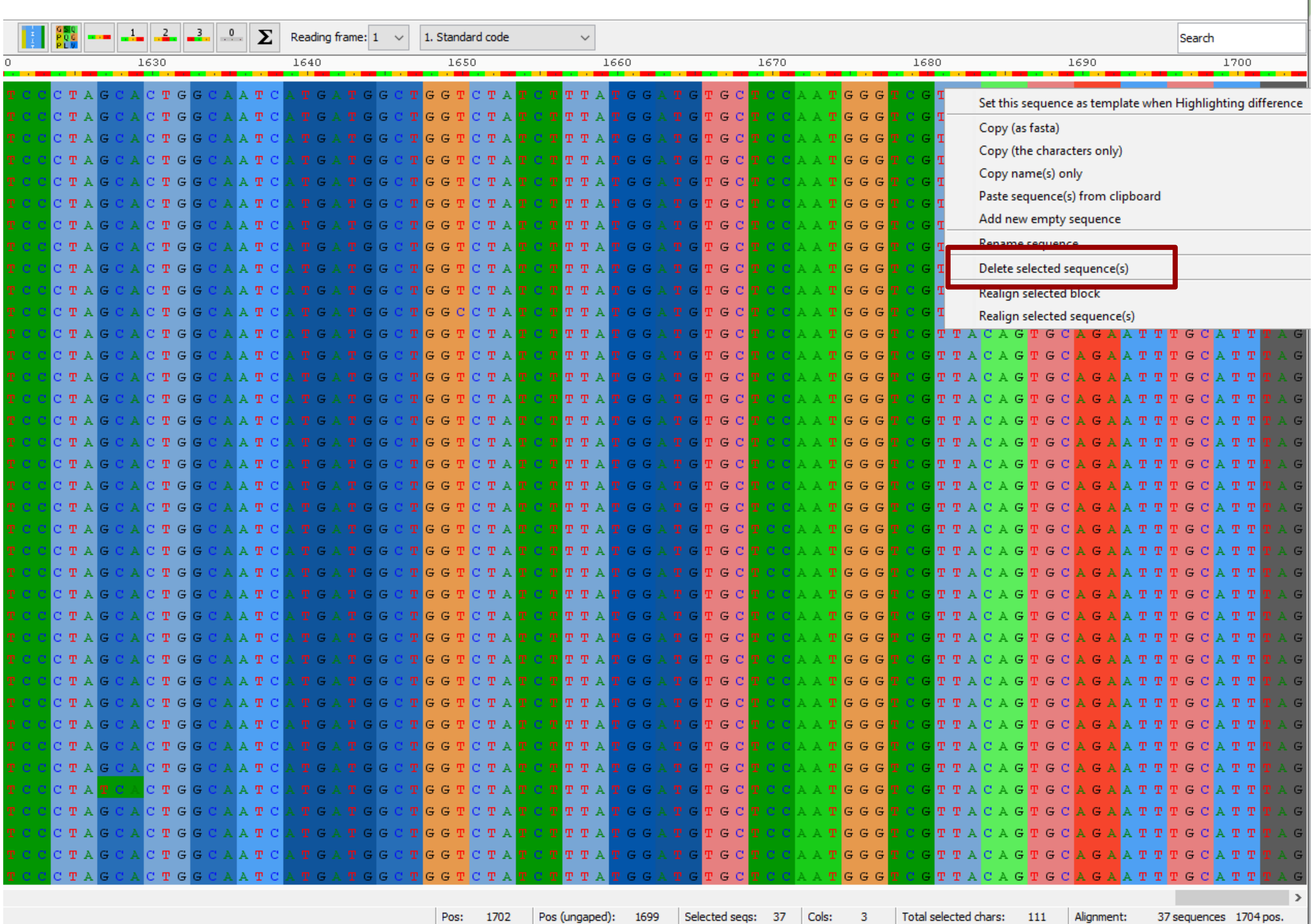




Stop codons are not modeled in selection analyses

Second letter

First letter	U	C	A	G	Third letter
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
U	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
U	UUA Leu	UCA Ser	**UAA Stop**	**UGA Stop**	A
U	UUG Leu	UCG Ser	**UAG Stop**	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
C	CUC Leu	CCC Pro	CAC His	CGC Arg	C
C	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
C	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
A	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
A	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
A	**AUG Met**	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
G	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
G	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
G	GUG Val	GCG Ala	GAG Glu	GGG Gly	G


Reading frame: 1
1. Standard code
Search
Set this sequence as template when Highlighting difference
Copy (as fasta)
Copy (the characters only)
Copy name(s) only
Paste sequence(s) from clipboard
Add new empty sequence
Rename sequence
Delete selected sequence(s)
Realign selected block
Realign selected sequence(s)
Pos: 1702
Pos (ungaped): 1699
Selected seqs: 37
Cols: 3
Total selected chars: 111
Alignment: 37 sequences 1704 pos.

Back to the phylogenetic tree

An official website of the United States government Here's how you know

NIH National Library of Medicine
National Center for Biotechnology Information

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NCBI Virus
Sequences for discovery

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Explore Virus Data

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Advanced Filters for GenBank Sequences | **Visual Filters for GenBank Sequences**

Selected Results: 36 | Align | **Build Phylogenetic Tree**

New! Randomized subsets in Downloads

You now have the option of downloading a smaller, randomized subset of the data shown in the Results table. Begin by using filters to refine your dataset, select the Nucleotide, Protein, or RefSeq Genome tab above the table for the datatype you would like to download, then follow the prompts in the Download menu. Our Help documentation has more information.

Refine Results — Reset

- Virus: Influenza A virus, taxid:11320
- Accession
- Sequence Length: **Min:** 1704 **Max:** 1704
- Ambiguous Characters
- Sequence Type
- RefSeq Genome Completeness
- Nucleotide Completeness

Applied Filters: Virus (1) · Genotype (1) · Geographic Region (1) · Proteins (1) · Sequence Length (1)

Nucleotide (36) | Protein (0) | RefSeq Genome (0) — Select Columns

Expand Table

Accession	Organism Name	Submitters	Organization	Release Date	Isolate	Species	Length	Nuc Completeness	Geo Location	Ho
OQ958756	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-005647-001	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958764	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-005647-002	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958772	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-009324-002	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958780	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-009324-009	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958788	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010200-002	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958796	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010200-003	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958804	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010321-001	Alphainfluenzavirus influ...	1704	partial	USA: New York	
OQ958812	Influenza A virus	Youk,S., et al.	United States Department...	2023-08-23	22-010321-002	Alphainfluenzavirus influ...	1704	partial	USA: New York	

Feedback

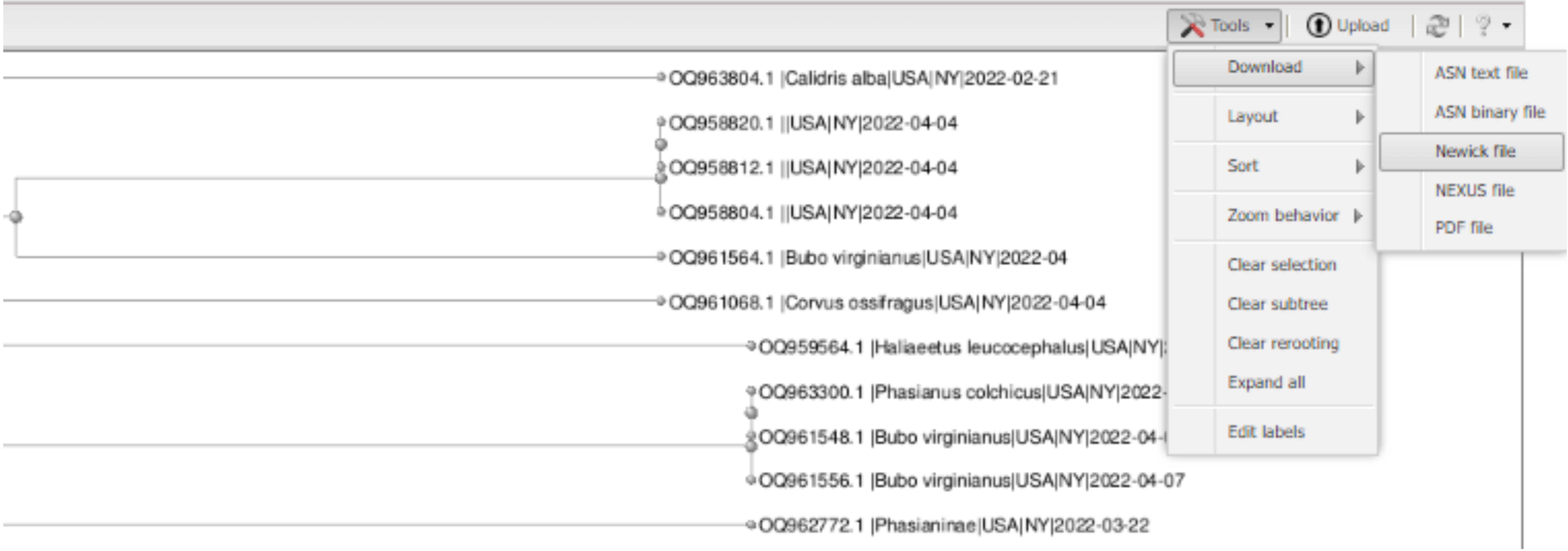
Phylogenetic Tree

Phylogenetic Tree

Phylogenetic Tree

Check for incomplete metadata or duplicate sequences which do not add evolutionary signal

Download the newick file



Create an FNA file

HyPhy uses a custom data format (Fasta + Newick or FNA) to make things easier

Simply open your Fasta file and paste the Newick string right below

```
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CATGAAACATCACTAGGGGTGAGCGCAGCTTGTCCATACCAGGGAGCGCCCTCCTTTTTC
AGAAATGTGGTGTGGCTTATCAAAAAGAACGATGCATACCCAACAATAAAGATAAGCTAC
AATAATACCAATCGGGAAGATCTCTTGATACTGTGGGGGATTCATCATTCCAACAATGCA
GAAGAGCAGACAAATCTCTACAAAAACCCAACCACCTACATTTCAGTTGGAACATCAACT
TTAAACCAGAGGTTGGCACCAAAAATAGCTACTAGATCCCAAGTAAACGGGCAACGTGGA
AGAATGGACTTCTTCTGGACAATCTTAAAACCAGATGATGCAATCCATTTCGAGAGTAAT
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```

With the Fasta and Newick files in hand…

You now have two files:

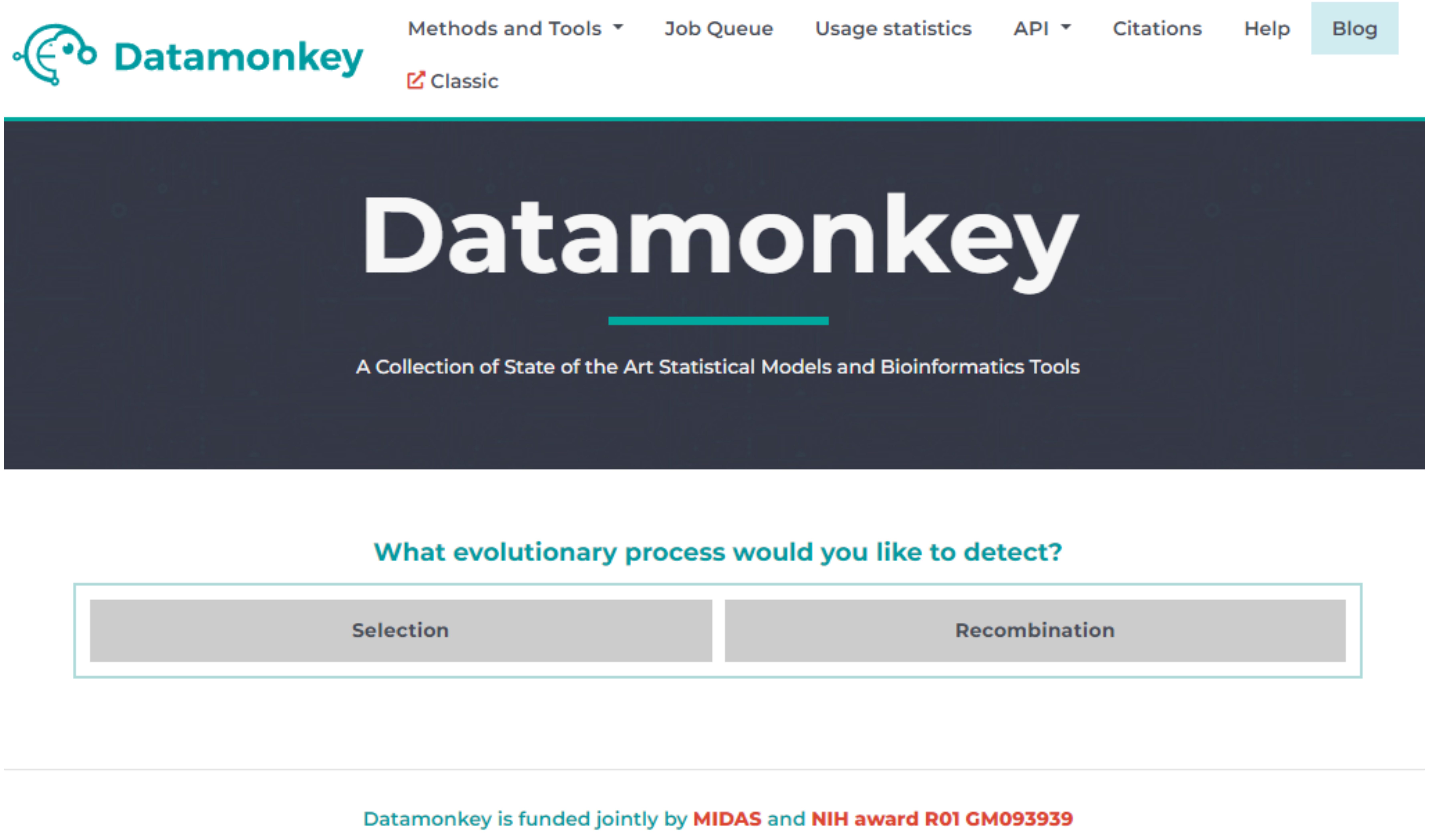
A codon-based multiple sequence alignment

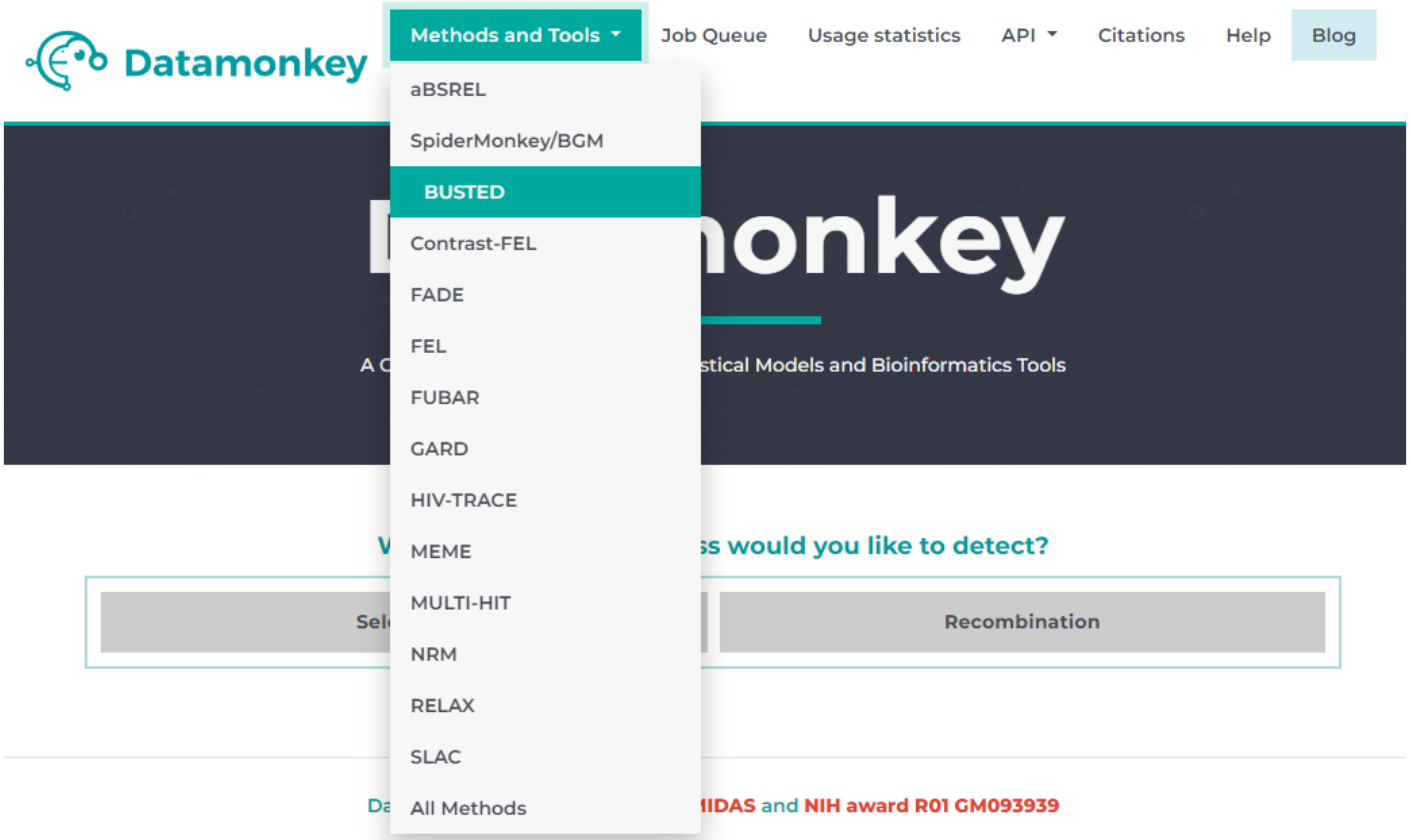
And a phylogenetic tree.

These can be used for all downstream selection analyses in HyPhy

Happy Hunting!

datamonkey.org will create a NJ-tree for your small dataset

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SpiderMonkey/BGM
BUSTED
Contrast-FEL
FADE
FEL
FUBAR
GARD
HIV-TRACE
MEME
MULTI-HIT
NRM
RELAX
SLAC
All Methods
stical Models and Bioinformatics Tools
Recombination

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Classic

BUSTED

Branch-site Unrestricted Statistical Test for Episodic Diversification

Choose File No file chosen

Genetic Code? Universal code

Synonymous rate variation (recommended)? Yes

Advanced Include support for multiple nucleotide substitutions?

Use standard models which permit only single nucleotide changes to occur instantly (Default)

NOTIFY WHEN COMPLETED

Email Address

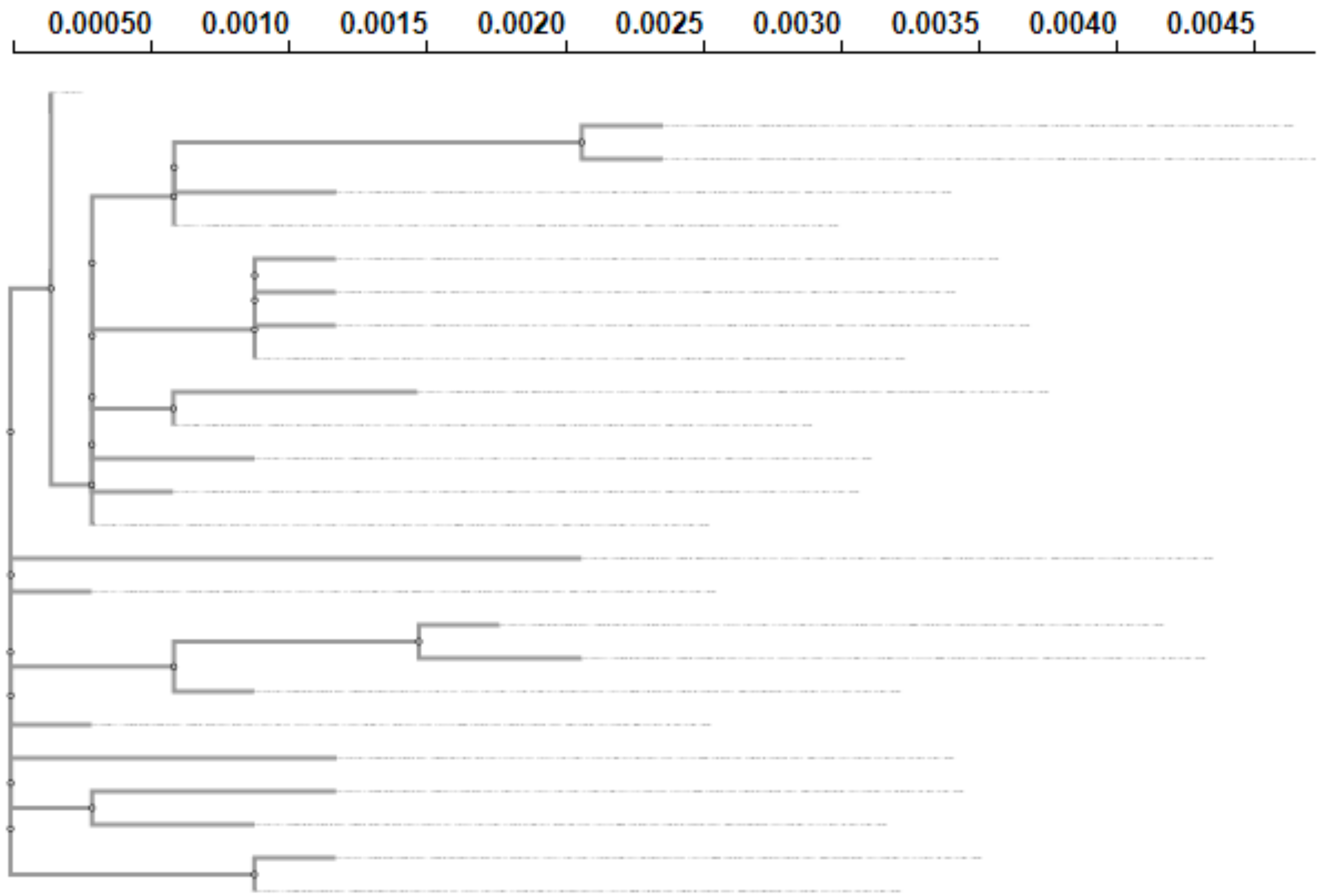
Run Analysis ▶

Datamonkey is funded jointly by MIDAS and NIH award R01 GM093939

BUSTED Branch-site Unrestricted Statistical Test for Episodic Diversification

Select Test Branches

Selection



Save Branch Selection

Datamonkey is funded jointly by MIDAS and NIH award R01 GM093939

BUSTED Branch-site Unrestricted Statistical Test for Episodic Diversification

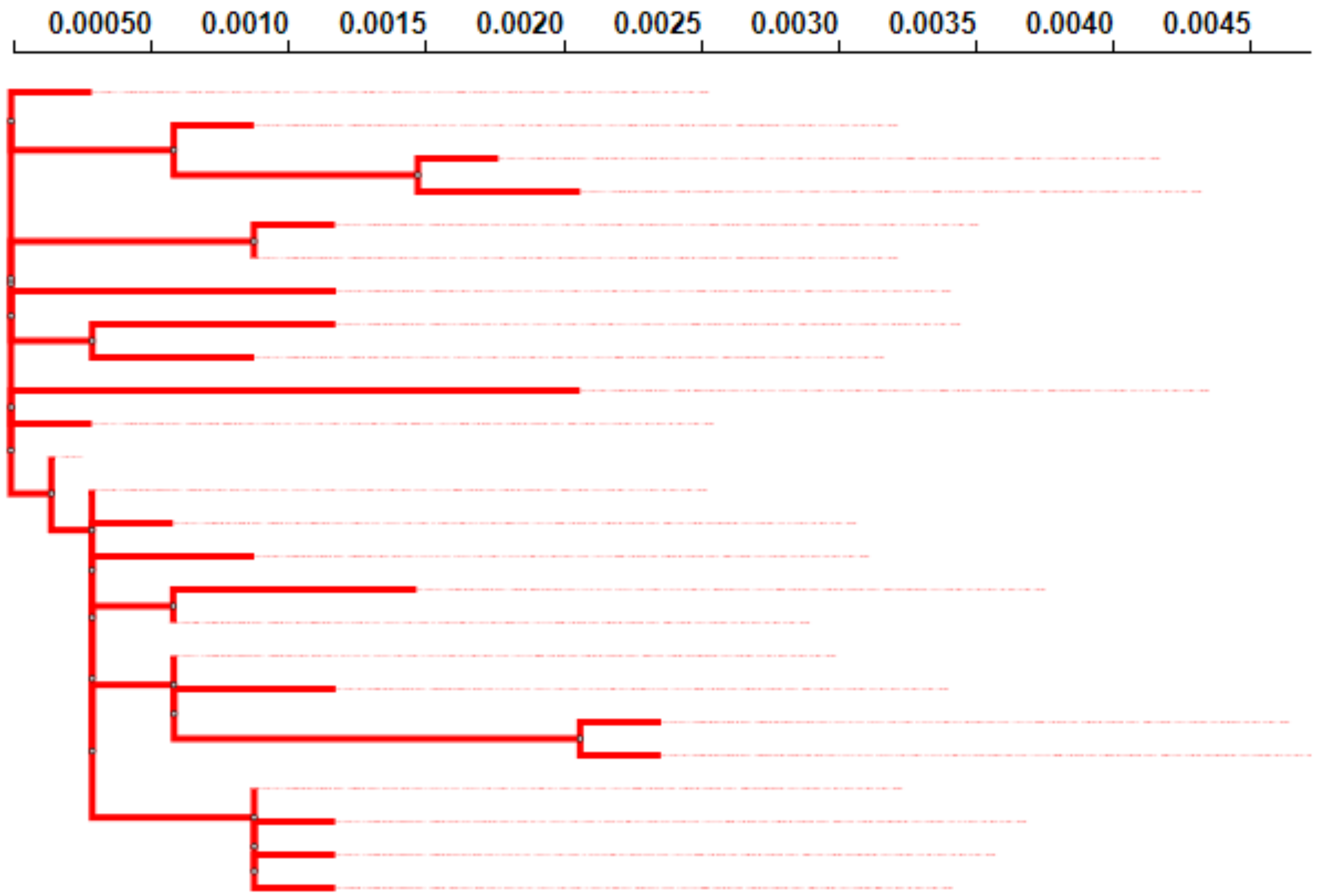
Select Test Branches

Datamonkey is funded jointly by MIDAS and NIH award R01 GM093939

BUSTED Branch-site Unrestricted Statistical Test for Episodic Diversification

Select Test Branches

Selection



Save Branch Selection

Datamonkey is funded jointly by MIDAS and NIH award R01 GM093939

Explore results - https://www.datamonkey.org/busted/66815fd6ab6e044dbc7ffd8a

BUSTED

Branch-site Unrestricted Statistical Test for Episodic Diversification

Based on the likelihood ratio test, there **is no** evidence of *episodic diversifying selection* in this dataset (p=0.5000).

Export

BUSTED analysis (v4.5) was performed on the alignment from /home/datamonkey/datamonkey-js-server/production/app/busted/output/66815fd6ab6e044dbc7ffd8a using HyPhy v2.5.61. This analysis **included** site-to-site synonymous rate variation.

Suggested citation: *Gene-wide identification of episodic selection*, Mol Biol Evol. 32(5):1365-71, *Synonymous Site-to-Site Substitution Rate Variation Dramatically Inflates False Positive Rates of Selection Analyses: Ignore at Your Own Peril*, Mol Biol Evol. 37(8):2430-2439

Evidence ratio threshold 10 Update

25 sequences in the alignment

567 codon sites in the alignment

1 partitions

47 median branches/partition used for testing

3 classes non-synonymous rate variation

3 classes synonymous rate variation

0.50 p-value for episodic diversityfing selection

0 Sites with ER≥10 for positive selection

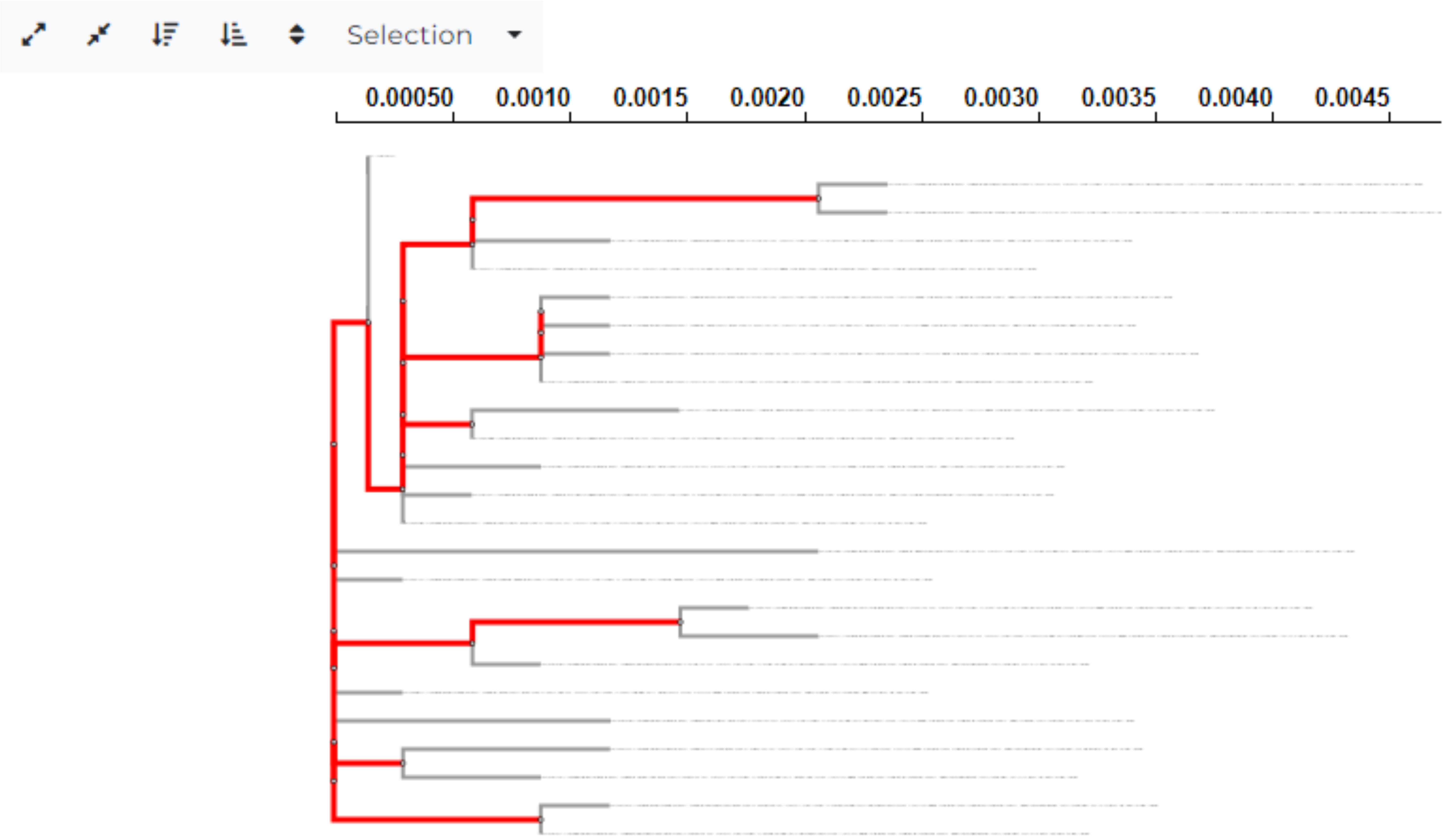
N/A:N/A Multiple hit rates (2H:3H)

Alignment-wide results

Model	Log (L)	AIC-c	Params.	Rate distribution	Rate plo
Unconstrained model	-2680.15	5503.02	71	**Tested ω** 0.1017 (100.00%) 0.08208 (0.0000%) 1.690 (0.0000%) Mean = 0.1017, CoV = 0.000	0.0 0.5
Unconstrained model				**Synonymous rates** 1.000 (100.00%) 1.179 (0.0000%) 2.673 (0.0000%) Mean = 1.000, CoV = 0.000	1.0 1.5 2.0

BUSTED Branch-site Unrestricted Statistical Test for Episodic Diversification

Select Test Branches



Save Branch Selection

Datamonkey is funded jointly by MIDAS and NIH award R01 GM093939

Explore results - https://www.datamonkey.org/busted/668160a0ab6e044dbc7ffdca

Evidence ratio threshold: 10 [Update]

25
sequences in the alignment

567
codon sites in the alignment

1
partitions

22
median branches/partition used for testing

3 classes
non-synonymous rate variation

3 classes
synonymous rate variation

0.50
p-value for episodic diversityfing selection

0
Sites with ER≥10 for positive selection

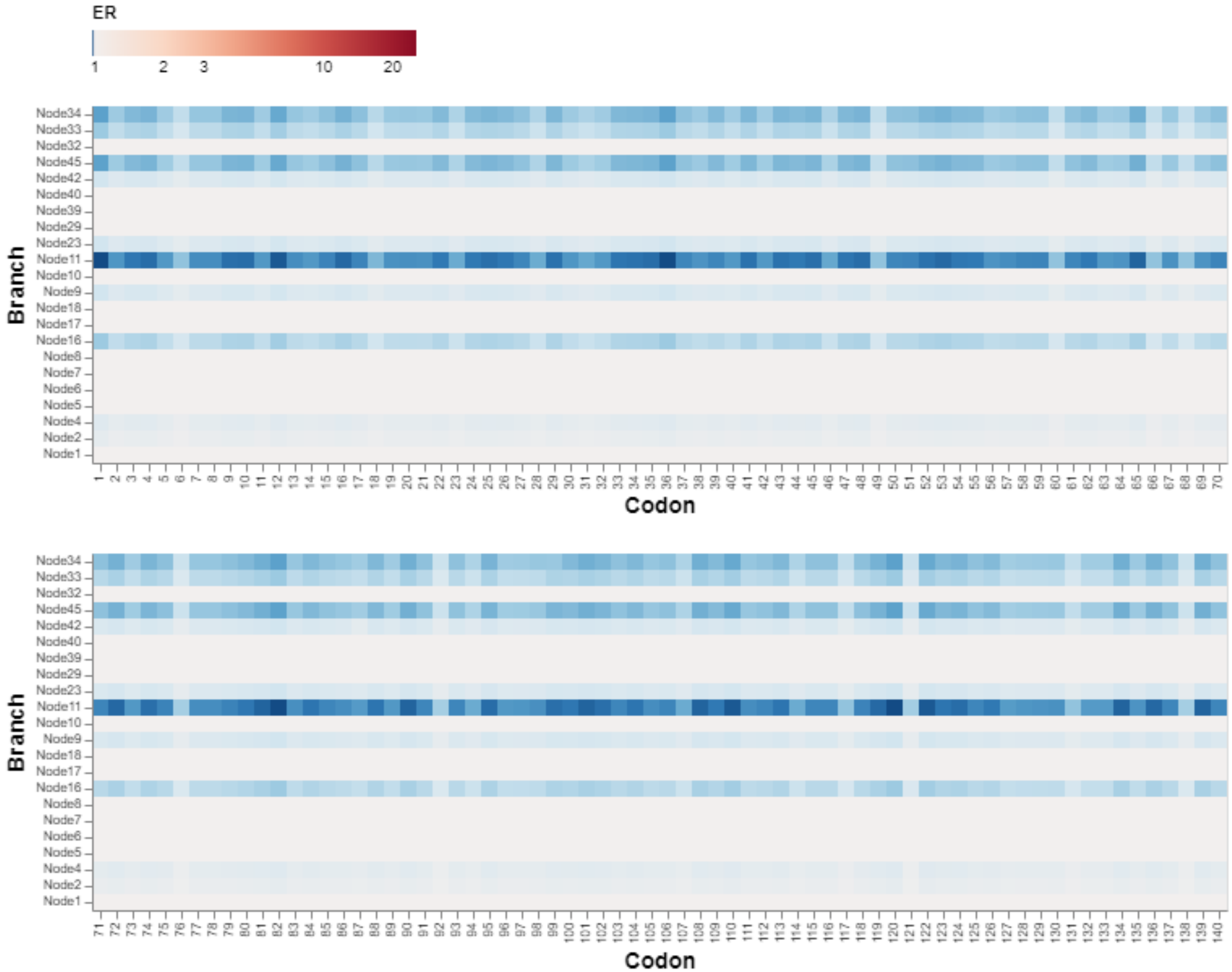
N/A:N/A
Multiple hit rates (2H:3H)

Alignment-wide results

Model	Log (L)	AIC-c	Params.	Rate distribution
Unconstrained model	-2679.26	5511.36	76	**Tested ω** 0.03200 (71.337%) 0.05756 (27.795%) 1.002 (0.86805%) Mean = 0.04752, CoV = 1.894
Unconstrained model				**Background ω** 0.08553 (24.916%) 0.1522 (75.084%) 0.4953 (0.0000%) Mean = 0.1356, CoV = 0.2126
Unconstrained model				**Synonymous rates** 1.000 (1.6086%) 1.000 (98.391%) 1.615 (0.0000000000000010924%) Mean = 1.000, CoV = 3.332e-8
Constrained model	-2679.26	5509.34	75	**Tested ω** 0.03280 (71.983%) 0.05772 (27.171%) 1.000 (0.84600%) Mean = 0.04775, CoV = 1.857
Constrained model				**Background ω** 0.08643 (25.064%) 0.1520 (74.936%) 0.4670 (0.0000%) Mean = 0.1356, CoV = 0.2097

Plot type Support for positive selection

Figure 1. Empirical Bayes Factors for ω>1 at a particular branch and site (only tested branches are included).



FEL

All branches

Fixed Effects Likelihood

FEL analysis was performed on the alignment from `/home/datamonkey/datamonkey-js-server/production/app/fel/output/668162d9ab6e044dbc7ffe34`. Statistical significance is evaluated based on 50 site-level parametric bootstrap replicates. This analysis **includes** site to site synonymous rate variation. Profile approximate confidence intervals for site-level dN/dS ratios have been computed.

Export

Suggested citation: Not So Different After All: A Comparison of Methods for Detecting Amino Acid Sites Under Selection (2005). *Mol Biol Evol* 22 (5): 1208-1222

p-value threshold 0.1 Update

25 sequences in the alignment

567 codon sites in the alignment

1 partitions

35 median branches/partition used for testing

55 non-invariant sites tested

50 parametric bootstrap replicates

0 Sites under diversifying positive selection at p≤0.1

24 Sites under purifying selection at p≤0.1

https://www.datamonkey.org/fel/668162d9ab6e044dbc7ffe34

FEL

Internal branches

Fixed Effects Likelihood

FEL analysis was performed on the alignment from `/home/datamonkey/datamonkey-js-server/production/app/fel/output/66816b85ab6e044dbc7ffe5a`. Statistical significance is evaluated based on 50 site-level parametric bootstrap replicates. This analysis **includes** site to site synonymous rate variation. Profile approximate confidence intervals for site-level dN/dS ratios have been computed.

Export

Suggested citation: Not So Different After All: A Comparison of Methods for Detecting Amino Acid Sites Under Selection (2005). *Mol Biol Evol* 22 (5): 1208-1222

p-value threshold 0.1 Update

25 sequences in the alignment

567 codon sites in the alignment

1 partitions

35 median branches/partition used for testing

46 non-invariant sites tested

50 parametric bootstrap replicates

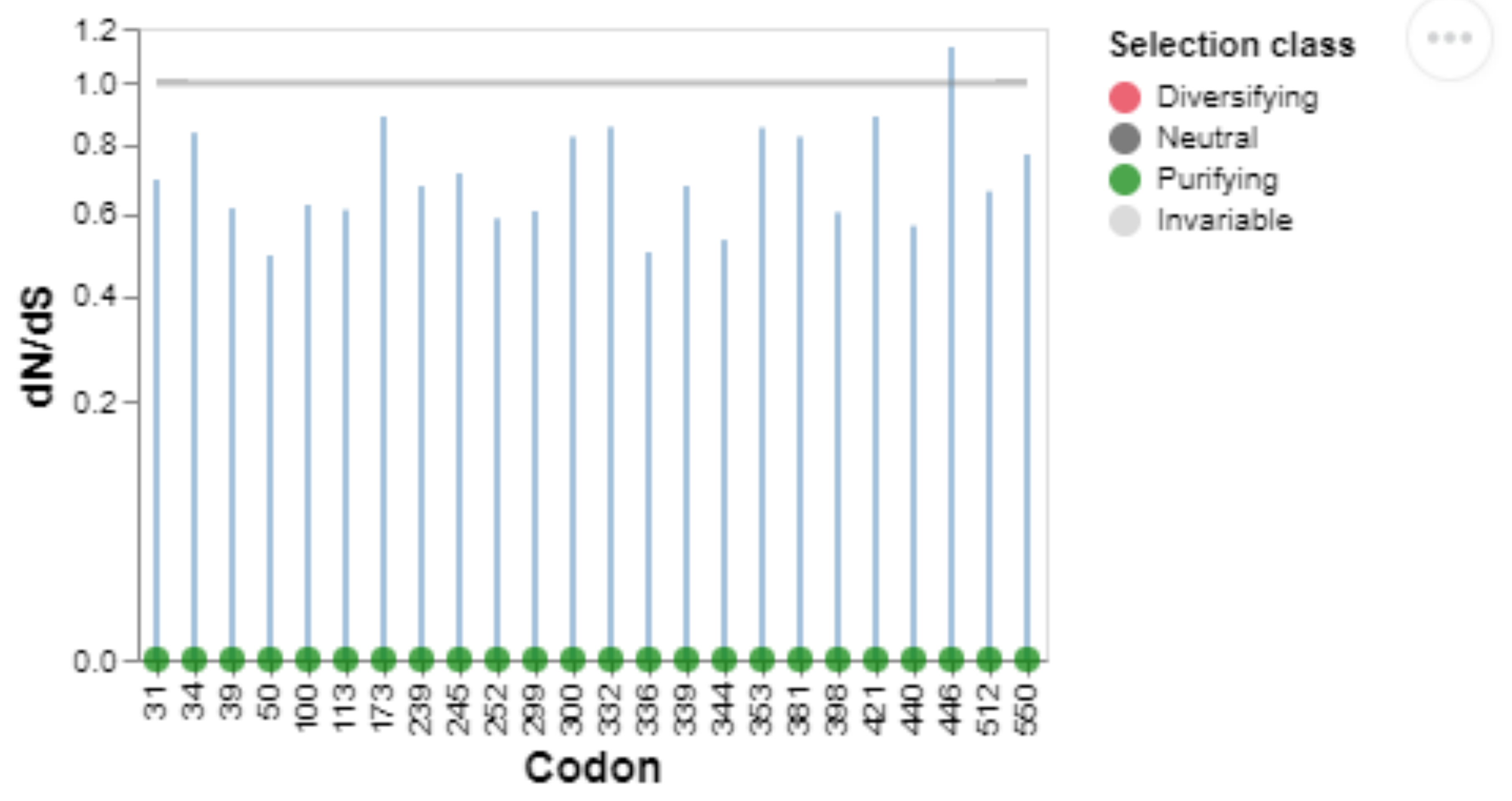
0 Sites under diversifying positive selection at p≤0.1

3 Sites under purifying selection at p≤0.1

https://www.datamonkey.org/fel/66816b85ab6e044dbc7ffe5a

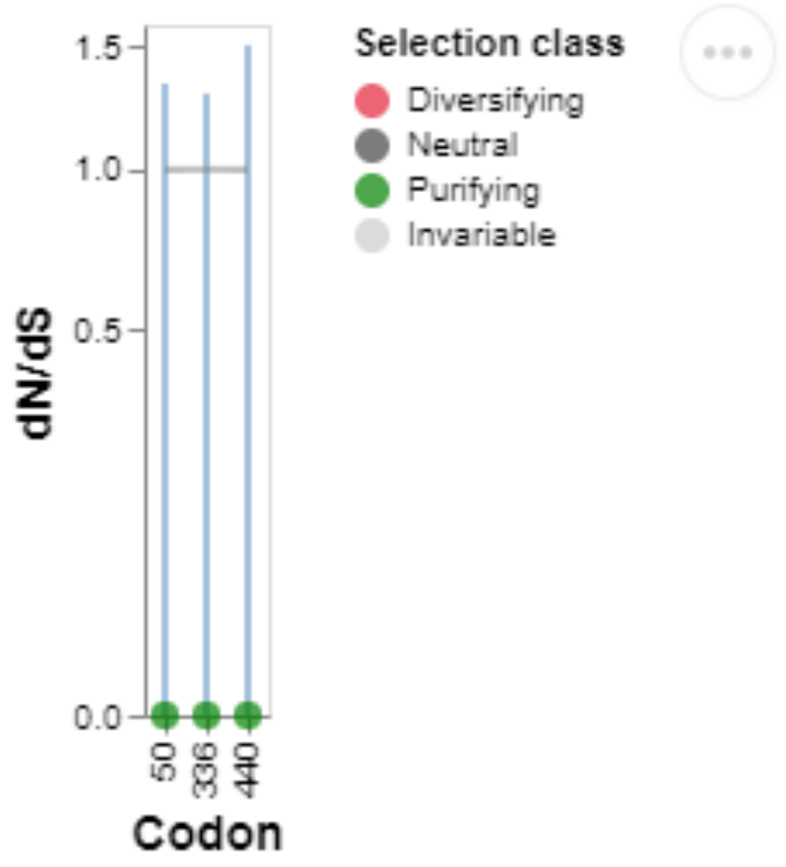
All branches

Figure 1. Maximum likelihood estimates of dN/dS at each site, together with estimated profile condifence intervals (if available). dN/dS = 1 (neutrality) is depicted as a horizontal gray line. Boundaries between partitions (if present) are shown as vertibal dashed lines.



Internal nodes only

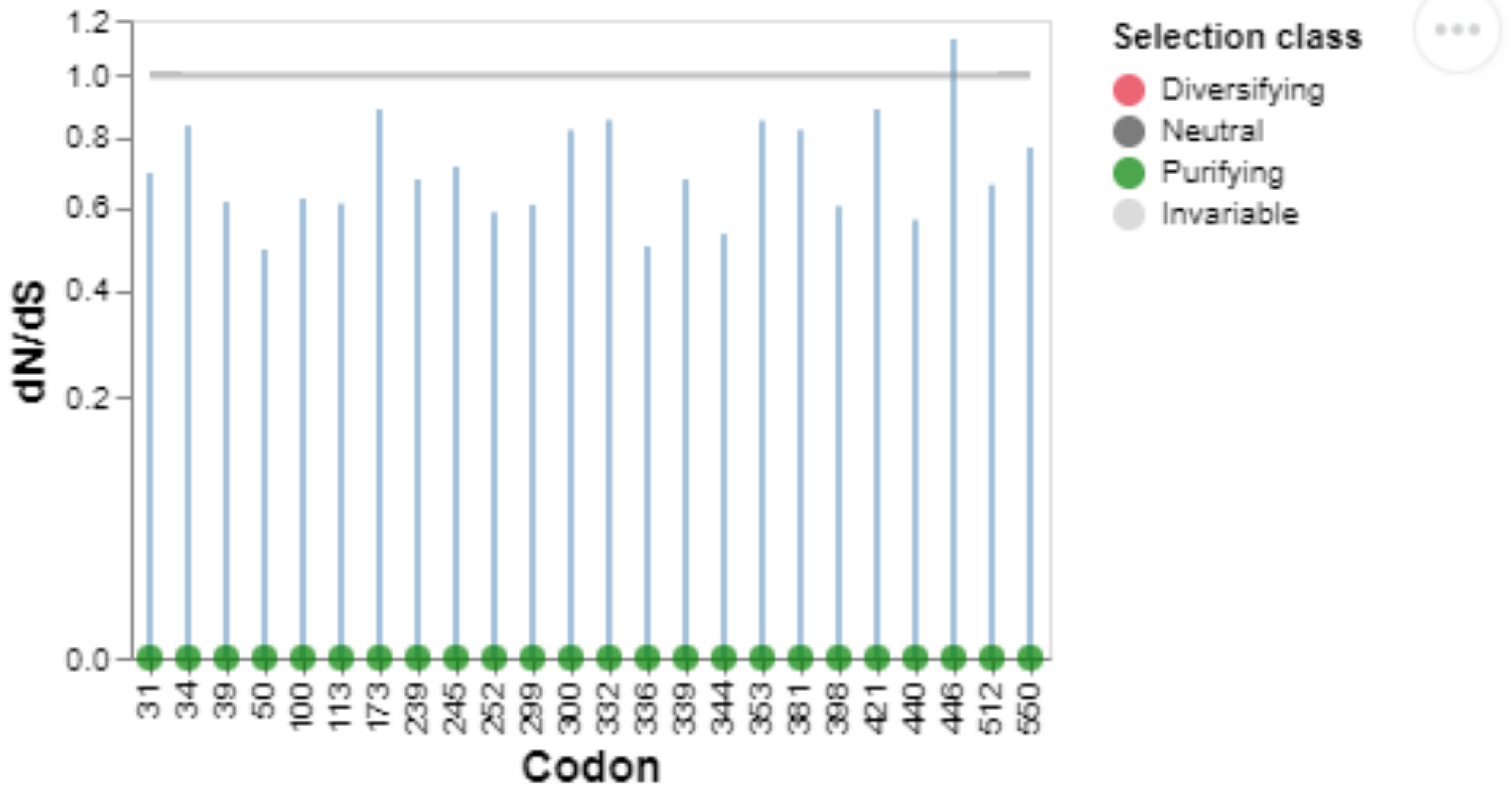
Figure 1. Maximum likelihood estimates of dN/dS at each site, together with estimated profile condifence intervals (if available). dN/dS = 1 (neutrality) is depicted as a horizontal gray line. Boundaries between partitions (if present) are shown as vertibal dashed lines.



Sites 50, 336 and 440 are found in both results

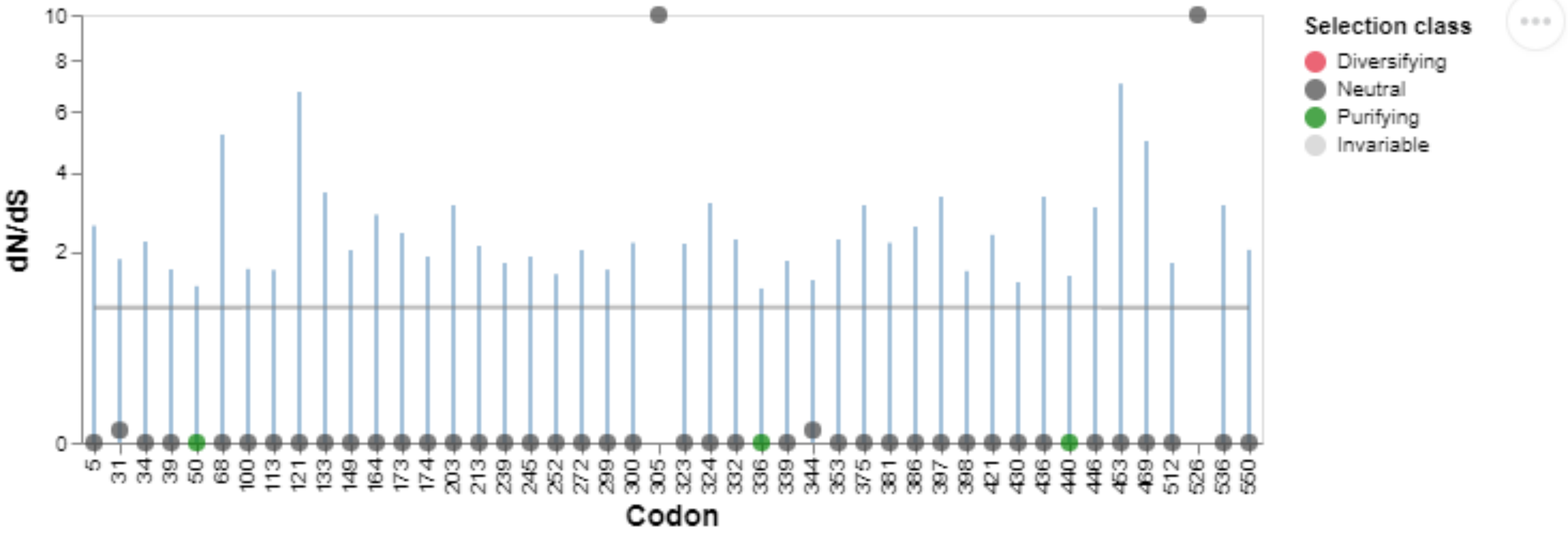
All branches

Figure 1. Maximum likelihood estimates of dN/dS at each site, together with estimated profile condifence intervals (if available). dN/dS = 1 (neutrality) is depicted as a horizontal gray line. Boundaries between partitions (if present) are shown as vertibal dashed lines.



Internal nodes only

Figure 1. Maximum likelihood estimates of dN/dS at each site, together with estimated profile condifence intervals (if available). dN/dS = 1 (neutrality) is depicted as a horizontal gray line. Boundaries between partitions (if present) are shown as vertibal dashed lines.



Sites 50, 336 and 440 are found in both results

Figure 1. Comparison of asymptotic vs boostrap LRT distributions (limited to 60 sites).

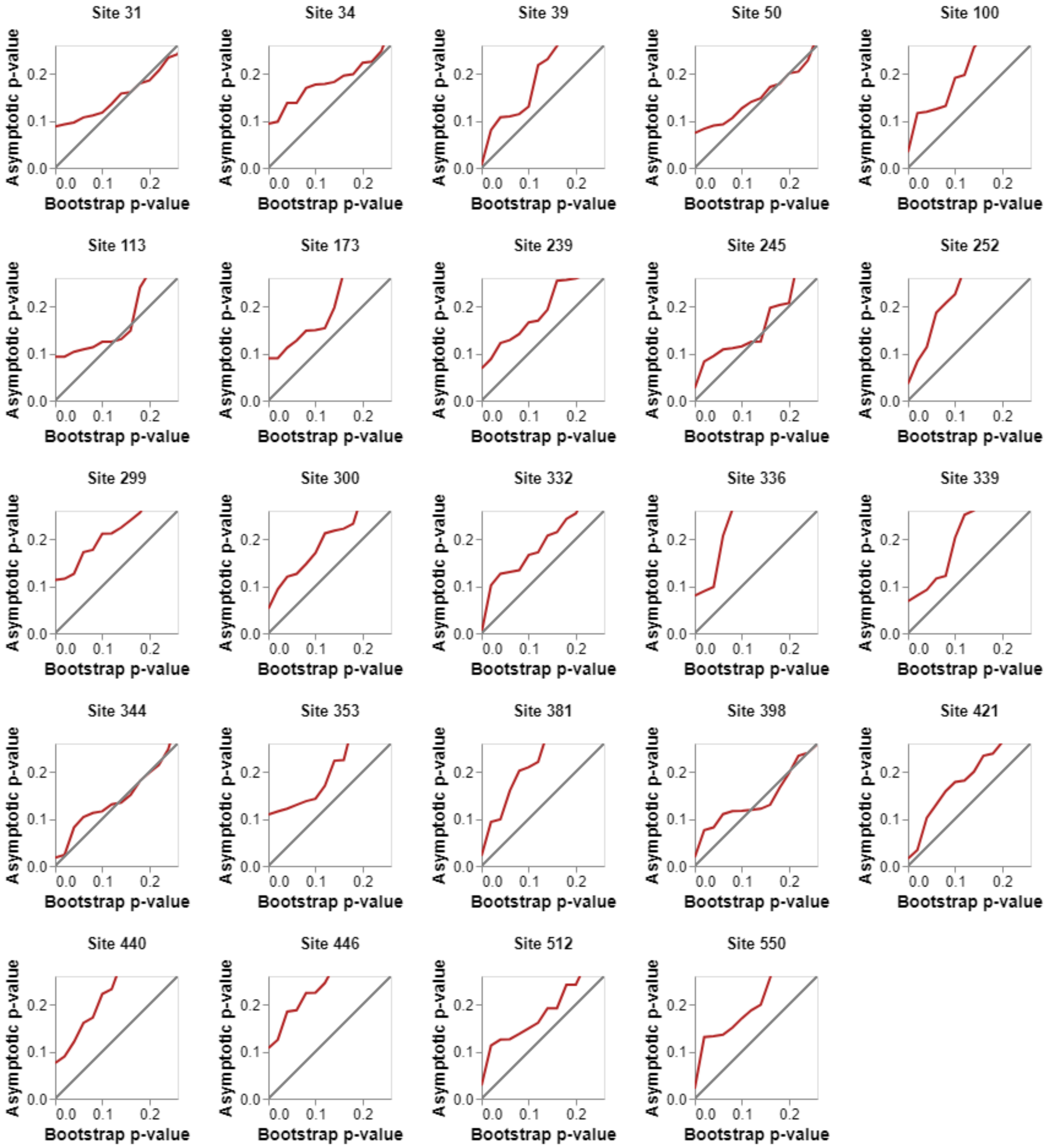


Figure 1. Comparison of asymptotic vs boostrap LRT distributions (limited to 60 sites).

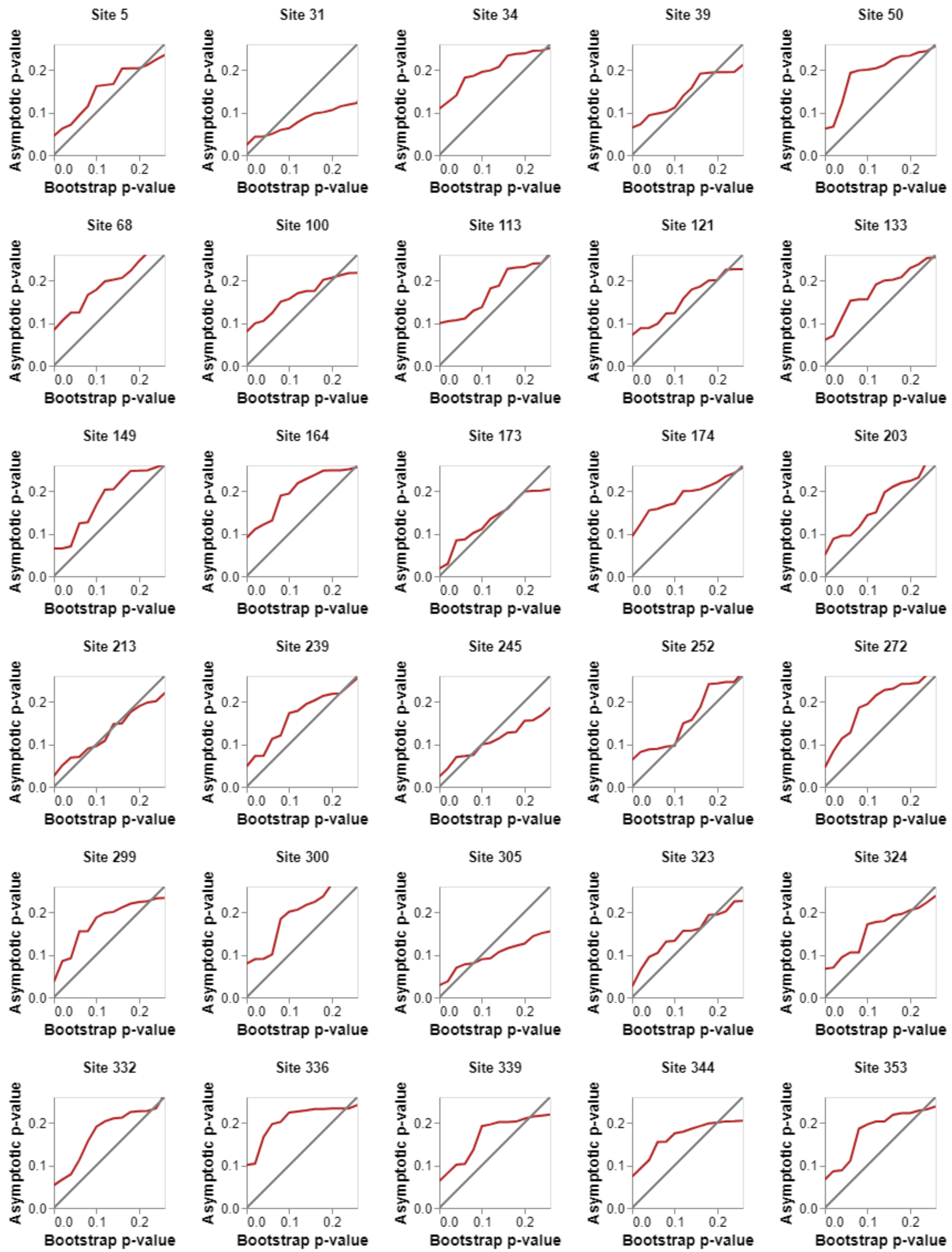


Figure 1. Comparison of asymptotic vs boostrap LRT distributions (limited to 60 sites).

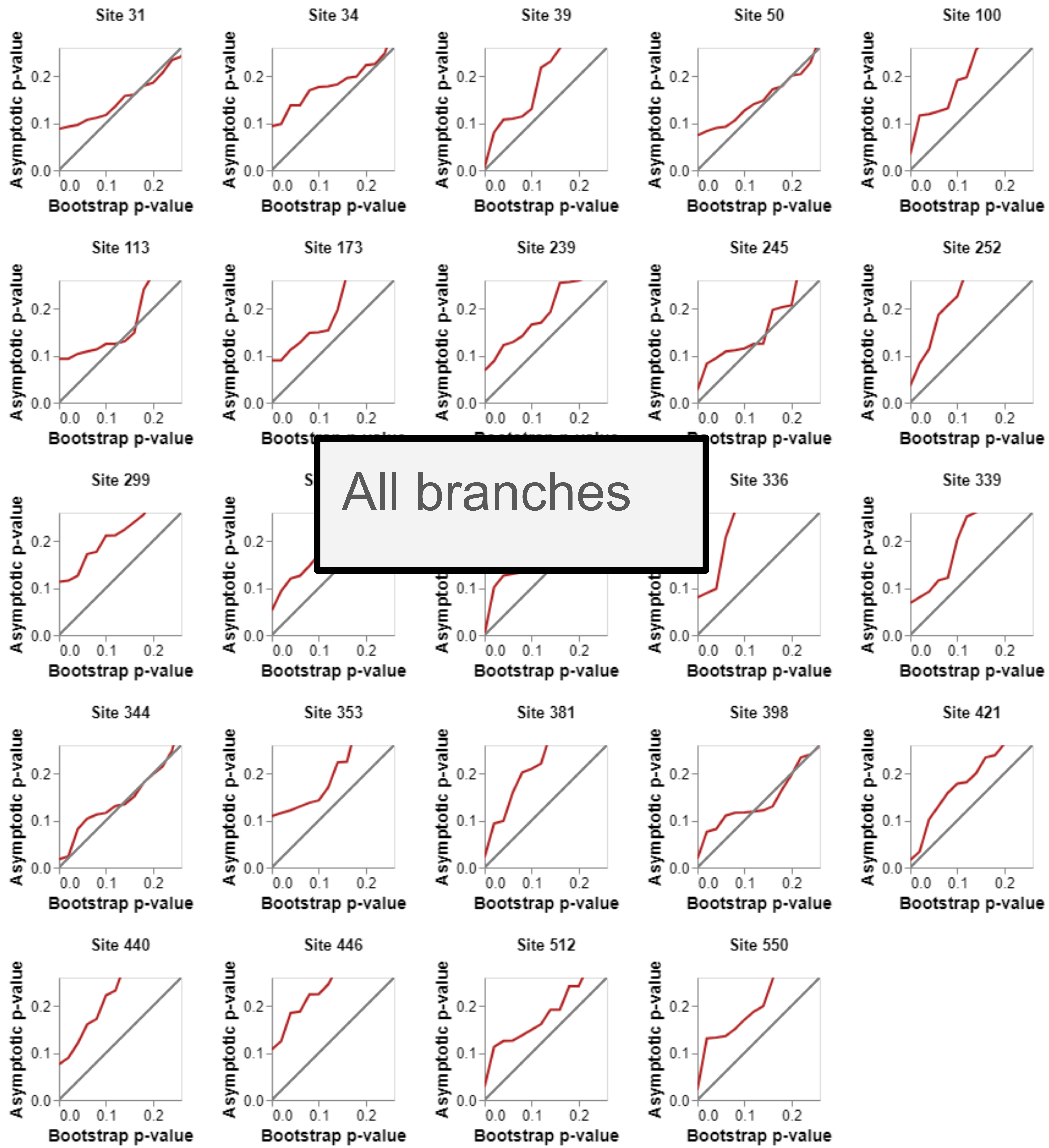
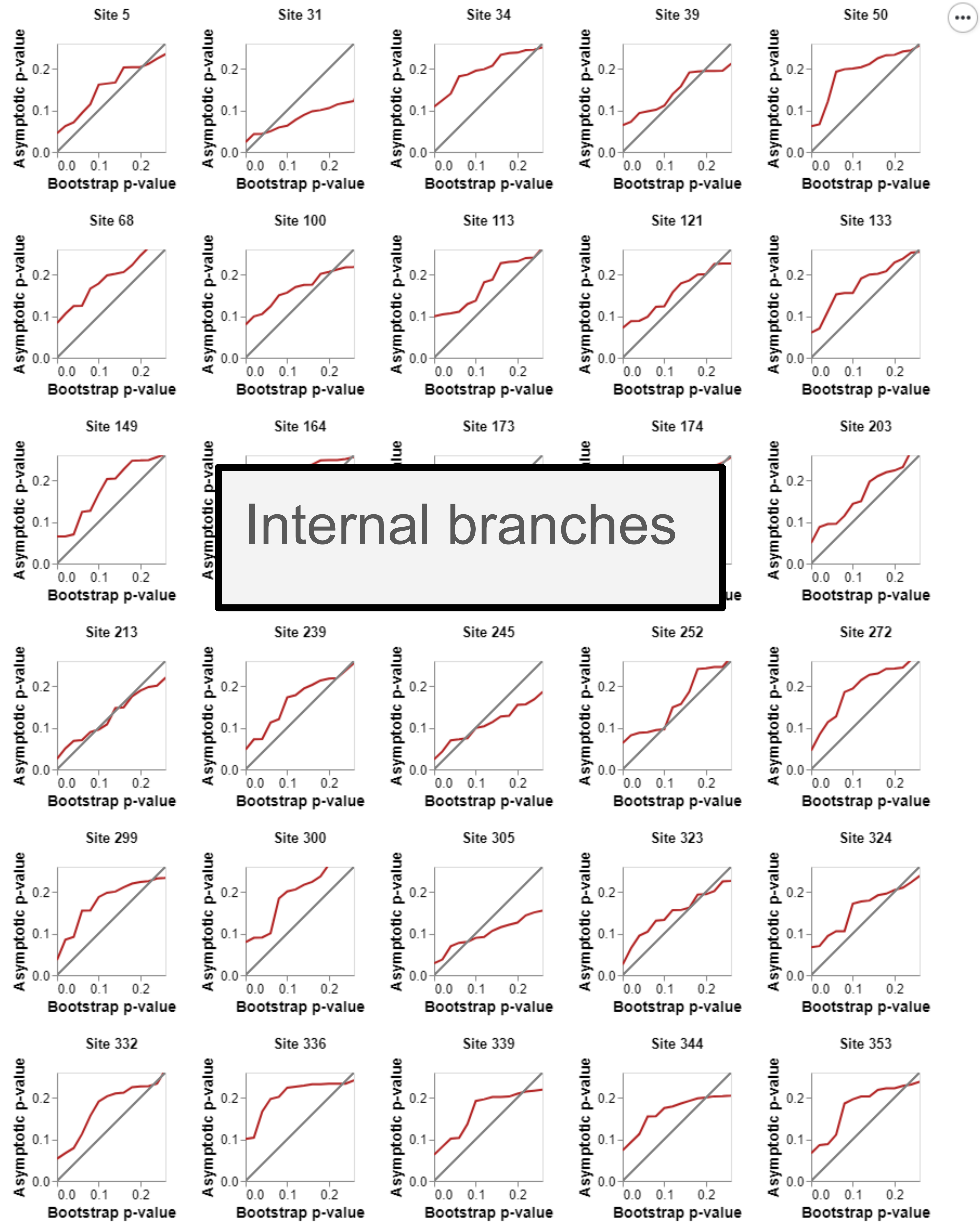


Figure 1. Comparison of asymptotic vs boostrap LRT distributions (limited to 60 sites).



All branches

Internal branches